

How to find all extremal graphs using symmetric subgraphs

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Abstract

Let $\mathcal{F}$ be a finite family of graphs with $\min_{F \in \mathcal{F}} \chi(F) = r + 1 \geq 3$, where $\chi(F)$ is the chromatic number of F . Set $t = \max_{F \in \mathcal{F}} |F|$. Let $\text{EX}(n, \mathcal{F})$ be the set of graphs with maximum edges among all the graphs of order n without any $F \in \mathcal{F}$ as a subgraph. Let $T(n, r)$ be the Turán graph of order n with r parts. Assume that some $F_0 \subseteq \mathcal{F}$ is a subgraph of the graph obtained from $T(rt, r)$ by embedding a path in its one part. Simonovits [20] introduced the concept of symmetric subgraphs, and proved that there exist graphs in $\text{EX}(n, \mathcal{F})$ which have symmetrical property. In this paper, we aim to find a way to characterize all the extremal graphs for such $\mathcal{F}$ using symmetric subgraphs. Some new extremal results are obtained.

Keywords: extremal graph; symmetric subgraph; the progressive induction; blow up of graphs.

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1 Introduction

All graphs considered in this paper are finite, undirected and without multi-edges or loops. We first introduce some elementary notations in graph theory. For a graph G , let $\overline{G}$ denote its complement. The vertex set and edge set of G are denoted by $V(G)$ and $E(G)$, respectively. Let $|G| = |V(G)|$ and $e(G) = |E(G)|$. For a subset $S \subseteq V(G)$, let $G[S]$ be the subgraph of G induced by S , and let $G - S = G[V(G) - S]$. For a vertex u , denote $G - \{u\}$ by $G - u$. Let $N_G(u)$ be the set of neighbors of u in G , and let $d_G(u) = |N_G(u)|$. Let $\delta(G)$ denote the minimum degree of G . For two disjoint subsets $U, S \subseteq V(G)$, let $e_G(U, S)$ be the number of edges between U and S . For any terminology used but not defined here, one may refer to [1].

For $\ell \geq 2$ graphs $G_1, G_2, \dots, G_\ell$, let $\cup_{1 \leq i \leq \ell} G_i$ (or $G_1 \cup G_2 \cup \dots \cup G_\ell$) denote the disjoint union of them. Let $\otimes_{1 \leq i \leq \ell} G_i$ (or $G_1 \otimes G_2 \otimes \dots \otimes G_\ell$) denote the graph obtained from

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$\cup_{1 \leq i \leq \ell} G_i$ by connecting each vertex in G_i to each vertex in G_j for any $1 \leq i \neq j \leq \ell$. For a positive integer t and a graph H , let $tH = \cup_{1 \leq i \leq t} H$. For a certain integer n , let K_n, C_n and P_n be the complete graph, the cycle and the path of order n , respectively. For $r \geq 2$, let $K_{n_1, n_2, \dots, n_r}$ be the complete r -partite graph with parts of sizes $n_1, n_2, \dots, n_r$. Let $T(n, r)$ be the Turán graph of order n with r parts (i.e., the complete r -partite graph of order n , in which each part has $\lfloor \frac{n}{r} \rfloor$ or $\lceil \frac{n}{r} \rceil$ vertices). Assume that $T(n, 1) = \overline{K_n}$, and $T(n, 0)$ is an empty graph. For a graph G , let $\chi(G)$ denote its *chromatic number*.

Let F be a graph. For a graph G , we say $F \subseteq G$ if G contains a copy of F , and G is F -free otherwise. Let $\mathcal{F}$ be a family of graphs. We say G is $\mathcal{F}$ -free, if G is F -free for any $F \in \mathcal{F}$. We also use F to denote $\mathcal{F}$, if $\mathcal{F} = \{F\}$. Let $\text{EX}(n, \mathcal{F})$ be the set of $\mathcal{F}$ -extremal graphs of order n (i.e., the $\mathcal{F}$ -free graphs of order n with maximum edges). Let $\text{ex}(n, \mathcal{F})$ be the number of edges of any graph in $\text{EX}(n, \mathcal{F})$. The classical Turán Theorem [22] states that $\text{EX}(n, K_{r+1}) = \{T(n, r)\}$, which is considered as the origin of extremal graph theory. Erdős, Stone and Simonovits [9, 10] proposed the stability theorem

$$\text{ex}(n, \mathcal{F}) = (1 - \frac{1}{r}) \frac{n^2}{2} + o(n^2),$$

where $\mathcal{F}$ is a finite family of graphs with $\min_{F \in \mathcal{F}} \chi(F) = r + 1 \geq 3$. The Turán type problem has attracted many researchers (for example, see [2, 3, 4, 5, 7, 8, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33]).

Simonovits [20] introduced the concept of symmetric subgraphs. Let G be a graph. For $\tau \geq 1$ induced subgraphs of G : $Q_1, Q_2, \dots, Q_\tau$, they are called symmetric subgraphs in G , if they are connected and vertex-disjoint, and there are isomorphisms $\psi_j : Q_1 \rightarrow Q_j$ for all $2 \leq j \leq \tau$ such that for any $u \in V(Q_1)$ and $v \in V(G) - (\cup_{1 \leq i \leq \tau} V(Q_i))$, $uv \in E(G)$ if and only if $\psi_j(u)v \in E(G)$. u and $\psi_j(u)$ are called corresponding vertices. These symmetric subgraphs are called symmetric vertices if $|Q_i| = 1$. Clearly, τ vertices $v_1, v_2, \dots, v_\tau$ of G are symmetric if and only if $\{v_1, v_2, \dots, v_\tau\}$ is independent and $N_G(v_1) = N_G(v_i)$ for any $1 \leq i \leq \tau$.

Observation 1. Let F be a graph of order $t \geq 2$. Let G be a F -free graph. Assume that $Q_1, Q_2, \dots, Q_t$ are symmetric subgraphs of G . Let G' be the graph obtained from G by adding a new copy of Q_1 , say Q , such that $Q, Q_1, Q_2, \dots, Q_t$ are symmetric subgraphs in G' . Then G' is also F -free. In fact, if $F \subseteq G'$, then $V(F) \cap V(Q) \neq \emptyset$ since G is F -free. Noting $t = |F|$, there is at least one of $Q_1, Q_2, \dots, Q_t$, say Q_t , such that $V(F) \cap V(Q_t) = \emptyset$. Let F' be the subgraph of G induced by $(V(F) - V(Q)) \cup V(Q_t)$. By symmetry of Q and Q_t in G' , we see that $F \subseteq F' \subseteq G$, a contradiction. Hence G' is F -free.

The above observation is very important in the study of extremal graphs using symmetric subgraphs. For certain integers n, r and c , let $\mathbb{D}(n, r, c)$ denote the family of graphs

G of order n satisfying the following symmetry condition:

- (i) It is possible to omit at most c vertices of G so that the remaining graph G' is of form $\otimes_{1 \leq i \leq r} G_i$, where $||V(G_i) - \frac{n}{r}|| \leq c$ for any $1 \leq i \leq r$.
- (ii) For any $1 \leq i \leq r$, $G_i = \cup_{1 \leq j \leq k_i} H_j^i$, where $H_1^i, H_2^i, \dots, H_{k_i}^i$ are symmetric subgraphs in G and $|H_1^i| \leq c$.

Simonovits [20] gave the following influential result on extremal graphs. Given integers r, t , when n is sufficiently large, $M := M(r, t)$ is called a constant if it is just a function of r and t (i.e., independent to n).

Theorem 1.1 ([20]) *Let $\mathcal{F}$ be a finite family of graphs with $\min_{F \in \mathcal{F}} \chi(F) = r + 1 \geq 3$. Set $t = \max_{F \in \mathcal{F}} |F|$. Assume that $F \subseteq P_t \otimes T(t(r-1), r-1)$ for some $F \in \mathcal{F}$. Then, for sufficiently large n , $\mathbb{D}(n, r, c)$ contains a graph G in $\text{EX}(n, \mathcal{F})$, where c is constant. Furthermore, if G is the only extremal graph in $\mathbb{D}(n, r, c)$, then it is unique in $\text{EX}(n, \mathcal{F})$.*

When $\mathbb{D}(n, r, c)$ contains only one $\mathcal{F}$ -free graph with maximum edges, then there is only one extremal graph for $\mathcal{F}$ by Theorem 1.1. We should also note that $\mathbb{D}(n, r, c)$ usually contains many extremal graphs. If so, Theorem 1.1 will not help us to find all extremal graphs. Motivated by this, we seek a way to characterize all extremal graphs in this paper. The following lemma is one main result of this paper. We will use it together with the progressive induction [19] to obtain some new extremal results.

Lemma 1.2 *Let $\mathcal{F}$ be a finite family of graphs with $\min_{F \in \mathcal{F}} \chi(F) = r + 1 \geq 3$. Set $t = \max_{F \in \mathcal{F}} |F|$. Assume that $F_1 \subseteq P_t \otimes T(t(r-1), r-1)$ and $F_2 \subseteq tK_{1,t} \otimes T(t(r-2), r-2)$ for some $F_1, F_2 \in \mathcal{F}$. For any given constant $N_0 := N_0(r, t)$, there exists a constant $O(r, t)$ such that any graph G in $\text{EX}(n, \mathcal{F})$ with n sufficiently large, has an induced subgraph $\otimes_{1 \leq i \leq r} G_i$ satisfying:*

- (i) $|G_i| = O(r, t) \cdot N_0$ for any $1 \leq i \leq r$;
- (ii) for any $1 \leq i \leq r$, $G_i = \cup_{1 \leq j \leq \ell_i} H_j^i$, where $H_1^i, H_2^i, \dots, H_{\ell_i}^i$ are symmetric subgraphs of G and $|H_1^i| \leq O(r, t)$.

For a finite family of graphs $\mathcal{F}$ with $\min_{F \in \mathcal{F}} \chi(F) = r + 1 \geq 3$ and $t = \max_{F \in \mathcal{F}} |F|$, define $q(\mathcal{F})$ to be the minimum integer s such that $F \subseteq \overline{K_s} \otimes T(tr, r)$ for some $F \in \mathcal{F}$. Clearly, $1 \leq q(\mathcal{F}) \leq t$.

Theorem 1.3 *Let $\mathcal{F}$ be a finite family of graphs with $\min_{F \in \mathcal{F}} \chi(F) = r + 1 \geq 3$. Set $t = \max_{F \in \mathcal{F}} |F|$ and $q = q(\mathcal{F})$. Assume that $F_1 \subseteq tK_2 \otimes T(t(r-1), r-1)$ and $F_2 \subseteq (K_{q-1,t} \cup K_{1,t}) \otimes T(t(r-1), r-1)$ for some $F_1, F_2 \in \mathcal{F}$. Then for sufficiently large n , any graph G in $\text{EX}(n, \mathcal{F})$ has a partition $V(G) = W \cup (\cup_{1 \leq i \leq r} S_i)$ satisfying:*

- (i) $|W| = q - 1$ and $|S_i| = \lfloor \frac{n-q+1}{r} \rfloor$ or $\lceil \frac{n-q+1}{r} \rceil$ for any $1 \leq i \leq r$;

(ii) for any $1 \leq i \leq r$, there exists $S'_i \subseteq S_i$ such that $|S_i - S'_i| \leq t^2$, and $N_G(v) = V(G) - S_i$ for any $v \in S'_i$.

Clearly, all the extremal graphs in Theorem 1.3 are contained in $\mathbb{D}(n, r, c)$ with $c = q + rt^2$. Note that in Theorem 1.3, the condition $F_2 \subseteq (K_{q-1,t} \cup K_{1,t}) \otimes T(t(r-1), r-1)$ for some $F_2 \in \mathcal{F}$ is not removable. For example, let $\mathcal{F} = \{3K_2 \otimes \overline{K_{10}}, P_5 \otimes \overline{K_{10}}\}$. When n is large, we can show that $G = H \otimes \overline{K_{\lfloor \frac{n}{2} \rfloor}}$ is an extremal graph for $\mathcal{F}$, where H is any double star. Clearly, when the two center vertices have almost equal degree in H , the resulting extremal graph G is not in $\mathbb{D}(n, r, c)$ for any constant c .

Simonovits (communicated with Liu [16]) asked the following question:

Question 1. Characterize graphs in $\mathcal{F}$ with $\min_{F \in \mathcal{F}} \chi(F) = r + 1 \geq 3$, such that $K_{q-1} \otimes T(n - q + 1, r)$ is the unique extremal graph.

Let I^{12} denote the icosahedron. Simonovits [21] observed that I^{12} has no independent set of order 4. Thus, $\chi(I^{12}) = 4$ and $K_2 \otimes T(n - 2, 3)$ is I^{12} -free. Moreover, $I^{12} \subseteq F_i$ for any $1 \leq i \leq 3$, where F_i is defined as in Theorem 1.4. As a classical result, Simonovits [21] characterized the unique extremal graph of large order for I^{12} , by showing the following stronger result.

Theorem 1.4 ([21]) *Let $\mathcal{F} = \{F_1, F_2, F_3\}$, where*

$$F_1 = P_6 \otimes 3K_1 \otimes 3K_1,$$

$$F_2 = (K_{1,2} \cup K_2) \otimes (K_2 \cup 2K_1) \otimes 3K_1,$$

$$F_3 = 2K_3 \otimes 3K_1 \otimes 3K_1.$$

Then $\text{EX}(n, \mathcal{F}) = \{K_2 \otimes T(n - 2, 3)\}$ for sufficiently large n . Consequently, $\text{EX}(n, I^{12}) = \{K_2 \otimes T(n - 2, 3)\}$ for sufficiently large n .

At the end of [21], Simonovits also hopes a generalized result. In this paper, we generalize Theorem 1.4 (by letting $\ell = 6, r = 3$) as follows, which supplies more examples for Question 1.

For certain integers n, r , let $K_{n_1, n_2, \dots, n_r}^+$ be the graph of order n obtained from the complete r -partite graph $K_{n_1, n_2, \dots, n_r}$ by adding a (near) perfect matching in each of its r parts, where $\sum_{1 \leq i \leq r} n_i = n$ and $|n_i - n_j| \leq 2$ for any $1 \leq i < j \leq r$. Let $\mathcal{G}_{n,r}$ be the set of all such graphs $K_{n_1, n_2, \dots, n_r}^+$. Let $G_{n,r}$ be the graph obtained from $T(n, r)$ by adding a (near) perfect matching in each of its r parts. Clearly, $G_{n,r}$ is an (but not only) extremal graph with maximum edges among the graphs in $\mathcal{G}_{n,r}$.

Theorem 1.5 Assume that $\ell \geq 4$ is an even integer, and r, m are two integers with $2 \leq r \leq \ell - 2$ and $m \geq \ell + 1$. Let $\mathcal{F} = \{F_1, F_2, F_3\}$, where

$$\begin{aligned} F_1 &= (P_\ell \cup \overline{K_m}) \otimes T(m(r-1), r-1), \\ F_2 &= (K_{1,2} \cup mK_2) \otimes mK_2 \otimes T(m(r-2), r-2), \\ F_3 &= (K_{\frac{\ell}{2}} \cup K_{\ell-1} \cup \overline{K_m}) \otimes T(m(r-1), r-1). \end{aligned}$$

Then for sufficiently large n ,

- (i). $\text{EX}(n, \mathcal{F}) = \left\{ K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r) \right\}$ for $2 \leq r \leq \ell - 3$;
- (ii). $\text{EX}(n, \mathcal{F}) \subseteq \left\{ K_{\frac{r}{2}} \otimes T(n - \frac{r}{2}, r) \right\} \cup \mathcal{G}_{n,r}$ for $2 \leq r = \ell - 2$.

Theorem 1.6 Assume that $m \geq 2$ is an integer. Then $\text{EX}(n, mI^{12}) = \{K_{3m-1} \otimes T(n - 3m + 1, 3)\}$ for sufficiently large n .

The following Theorem 1.7 generalizes the extremal results for the p -th power of paths [24, 27] and the odd prism [15].

Theorem 1.7 For $\ell \geq 3$ and $m \geq 2\ell$, let $\mathcal{F} = \{F_1, F_2, F_3, F_4\}$, where

$$\begin{aligned} F_1 &= (P_\ell \cup \overline{K_m}) \otimes T(m(r-1), r-1), \\ F_2 &= mK_2 \otimes mK_2 \otimes T(m(r-2), r-2), \\ F_3 &= (2K_{\ell-1} \cup \overline{K_m}) \otimes (K_2 \cup \overline{K_m}) \otimes T(m(r-2), r-2), \\ F_4 &= (K_{\ell-1} \cup \overline{K_m}) \otimes (K_{\ell-1} \cup \overline{K_m}) \otimes T(m(r-2), r-2). \end{aligned}$$

For sufficiently large n , any graph $G \in \text{EX}(n, \mathcal{F})$ satisfies $G = H \otimes T(n - |H|, r-1)$, where H is an extremal P_ℓ -free graph.

The rest of this paper is organized as follows. In Section 2, we apply our results on the extremal problems for blow up or odd-ballooning of graphs. In Section 3, we include some lemmas needed in the proofs of the main results. In Section 4, we give the proof of Lemma 1.2. In Section 5, we give the proof of Theorem 1.3. In Section 6, we give the proofs of Theorem 1.5 and Theorem 1.6. The proof of Theorem 1.7 is given in Section 7.

2 Applications on extremal graphs for blow up or odd-ballooning of graphs

Let $\mathcal{F}$ be a finite family of graphs with $\min_{F \in \mathcal{F}} \chi(F) = r + 1 \geq 3$. Let $\mathcal{M}(\mathcal{F})$ be the family of minimal graphs (with no isolated vertices) M satisfying the following: there exist an $F \in \mathcal{F}$ such that $F \subseteq (M \cup \overline{K_t}) \otimes T((r-1)t, r-1)$, where $t = |F|$. We call $\mathcal{M}(\mathcal{F})$ the decomposition family of $\mathcal{F}$. The decomposition family $\mathcal{M}(\mathcal{F})$ always contains some bipartite graphs.

A covering of a graph is a set of vertices incident with all edges of the graph. An independent covering of a bipartite graph is an independent set incident with all edges. Let $\beta(F)$ denote the covering number of F , i.e., the minimum number of vertices in a covering of F . If F is bipartite, let $q(F)$ be the minimum order of an independent covering of F . Let $\mathcal{F}$ be a finite family of graphs containing at least one bipartite graph. Define $q(\mathcal{F}) = \min \{q(F) \mid F \in \mathcal{F} \text{ is bipartite}\}$. This definition of $q(\mathcal{F})$ is consistent with the one in the introduction. The subgraph covering family $\mathcal{B}(\mathcal{F})$ of $\mathcal{F}$ is the set of subgraphs $F[U]$ of $F \in \mathcal{F}$, where U is a covering of F with $|U| \leq q(\mathcal{F}) - 1$. If $\beta(F) \geq q(F)$ for each $F \in \mathcal{F}$, set $\mathcal{B}(\mathcal{F}) = \{K_{q(\mathcal{F})}\}$.

The following result is an easy consequence of Theorem 1.3. (There is no edge inside S_i in the condition that $F_2 \subseteq K_{q-1,t} \cup K_2$ with some $F_2 \in \mathcal{M}(\mathcal{F})$.)

Corollary 2.1 *Let $\mathcal{F}$ be a finite family of graphs with $\min_{F \in \mathcal{F}} \chi(F) = r + 1 \geq 3$. Set $t = \max_{F \in \mathcal{F}} |F|$ and $q = q(\mathcal{F})$. Assume that $F_1 \subseteq tK_2, F_2 \subseteq K_{q-1,t} \cup K_2$ for some $F_1, F_2 \in \mathcal{M}(\mathcal{F})$. Then for sufficiently large n , any graph $G \in \text{EX}(n, \mathcal{F})$ is of form $G = H \otimes T(n - q + 1, r)$, where $H \in \text{EX}(q - 1, \mathcal{B}(\mathcal{F}))$.*

Corollary 2.1 has concrete examples (see [3, 28]). Given a graph G and an integer p , the blow-up of G , denoted by G^{p+1} , is the graph obtained from G by replacing each edge in G by a clique of size $p + 1$, where the new vertices of the cliques are all distinct. Liu [16] proved the following result.

Theorem 2.2 ([16]) *For any $p \geq 2$ and $k \geq 3$, when n is sufficiently large, we have the following results:*

- (i) *For any $C_k^{p+1} \neq C_3^3$, the unique graph in $\text{EX}(n, C_k^{p+1})$ is $K_{\lfloor \frac{k-1}{2} \rfloor} \otimes T(n - \lfloor \frac{k-1}{2} \rfloor, p)$ for odd k ; the unique graph in $\text{EX}(n, C_k^{p+1})$ is obtained from $K_{\lfloor \frac{k-1}{2} \rfloor} \otimes T(n - \lfloor \frac{k-1}{2} \rfloor, p)$ by adding one edge in one part of $T(n - \lfloor \frac{k-1}{2} \rfloor, p)$ for even k .*
- (ii) *For C_3^3 , if $4 \mid n$, the graph G_n^* (obtained from $T(n, 2)$ by adding a (near) perfect matching in each of its two parts) is the unique graph in $\text{EX}(n, C_3^3)$; otherwise both G_n^* and $K_1 \otimes T(n - 1, 2)$ are in $\text{EX}(n, C_3^3)$.*

For the extremal graphs for C_3^3 , one can see that

$$C_3^3 \subseteq P_4 \otimes \overline{K_2}, K_{1,2} \otimes (K_2 \cup K_1), K_3 \otimes \overline{K_3}.$$

From (ii) of Theorem 1.5 (by letting $r = 2, \ell = 4$), we can obtain $\text{EX}(n, C_3^3) \subseteq \{K_1 \otimes T(n-1, 2)\} \cup \mathcal{G}_{n,2}$. Thus, we have the following result (after a simple calculation).

Corollary 2.3 *For C_3^3 and large n , we have*

$$\begin{aligned} \text{EX}(n, C_3^3) &= \left\{ K_{\frac{n}{2}, \frac{n}{2}}^+ \right\} \text{ for } n \equiv 0 \pmod{4}; \\ \text{EX}(n, C_3^3) &= \left\{ K_{\frac{n-1}{2}, \frac{n+1}{2}}^+, K_1 \otimes T(n-1, 2) \right\} \text{ for } n \equiv 1, 3 \pmod{4}; \\ \text{EX}(n, C_3^3) &= \left\{ K_{\frac{n}{2}, \frac{n}{2}}^+, K_{\frac{n-2}{2}, \frac{n+2}{2}}^+, K_1 \otimes T(n-1, 2) \right\} \text{ for } n \equiv 2 \pmod{4}. \end{aligned}$$

For a general graph F , the extremal graphs for F^{p+1} with $p \geq \chi(F) + 1$ have a specified structure as follows (using Theorem 1.3 and a characterization for $\mathcal{M}(F^{p+1})$ in [16]).

Theorem 2.4 *Let F be a graph, and let $p \geq \chi(F) + 1$ be an integer. Set $q = q(\mathcal{M}(F^{p+1}))$. Then for sufficiently large n , any graph G in $\text{EX}(n, F^{p+1})$ has a partition $V(G) = W \cup (\cup_{1 \leq i \leq p} S_i)$ satisfying:*

- (i) $|W| = q - 1$ and $|S_i| = \lfloor \frac{n-q+1}{p} \rfloor$ or $\lceil \frac{n-q+1}{p} \rceil$ for any $1 \leq i \leq p$;
- (ii) there is a constant $M := M(|F|, p)$, such that for any $1 \leq i \leq p$, there exists $S'_i \subseteq S_i$ such that $|S_i - S'_i| \leq M$, and $N_G(v) = V(G) - S_i$ for any $v \in S'_i$.

Note that in Theorem 2.4, when F is non-bipartite, the graphs in $\text{EX}(n, F^{p+1})$ (with $p \geq \chi(F) + 1$) have been completely characterized by Yuan [28]. The extremal graphs for F^{p+1} are also characterized when F is a tree (see [3, 12, 16, 23]) and other specified graphs ([17]).

For a graph F , the odd-ballooning of F is a graph obtained from F , such that each edge in F is replaced by an odd cycle and all new vertices for different odd cycles are distinct. It seems that F is more concerned when it is a bipartite graph (see [13, 14, 29, 30, 31, 32]). Let $F = F[A, B]$ with $|A| \leq |B|$ be a bipartite graph. An odd-ballooning of F is called good, if the replacing triangles can exist only when the corresponding edges are incident with a leaf in B . Let F^{good} be a good odd-ballooning of F . One can check that $\mathcal{M}(F^{\text{good}})$ satisfies the condition in Theorem 1.3. Thus, we also have a specified characterization.

Theorem 2.5 *Let $F = F[A, B]$ with $|A| \leq |B|$ be a bipartite graph, and let F^{good} be a good odd-ballooning of it. Set $q = q(\mathcal{M}(F^{\text{good}}))$. Then for sufficiently large n , any graph G in $\text{EX}(n, F^{\text{good}})$ has a partition $V(G) = W \cup S_1 \cup S_2$ satisfying:*

- (i) $|W| = q - 1$ and $|S_i| = \lfloor \frac{n-q+1}{2} \rfloor$ or $\lceil \frac{n-q+1}{2} \rceil$ for any $1 \leq i \leq 2$;
- (ii) there is a constant $M := M(|F^{\text{good}}|)$, such that for any $1 \leq i \leq 2$, there exists $S'_i \subseteq S_i$ such that $|S_i - S'_i| \leq M$, and $N_G(v) = V(G) - S_i$ for any $v \in S'_i$.

Let $T = T[A, B]$ with $|A| \leq |B|$ be a tree. Define $\delta(A) = \min_{u \in A} d_T(u)$. Let T_0 be a good odd-ballooning of T . Let $\delta_1(A)$ be the minimum number of triangles containing a vertex $v \in A$ satisfying $d_T(v) = \delta(A)$. Using Theorem 1.1, Zhu and Chen [31] obtained $\text{ex}(n, T_0)$ and an extremal graph for any tree T . When $\delta_1(A) = \delta(A)$, then T_0 is just the friendship graph (see [8]). When $\delta_1(A) < \delta(A) = 1$, all the extremal graphs can be obtained by Corollary 2.1. When $\delta_1(A) < \delta(A)$ and $\delta(A) \geq 2$, their result [31] is the following. For certain integers n, a, k , let $G_{n,a,k}$ be the graph obtained from $K_{a-1} \otimes T(n - a + 1, 2)$ by adding a $K_{k-1,k-1}$ in one part of $T(n - a + 1, 2)$.

Theorem 2.6 ([31]) *Let $T = T[A, B]$ be a tree with $a = |A| \leq |B|$ and $\delta(A) = k \geq 2$, and let T_0 be a good odd-ballooning of T such that $\delta_1(A) < \delta(A)$. If n is sufficiently large, then*

$$\text{ex}(n, T_0) = \binom{a-1}{2} + (a-1)(n-a+1) + e(T(n-a+1, 2)) + (k-1)^2.$$

Furthermore, $G_{n,a,k}$ is an extremal graph.

Let $G'_{n,a,4}$ be the graph obtained from $K_{a-1} \otimes T(n - a + 1, 2)$ by adding a $3K_3$ in one part of $T(n - a + 1, 2)$. Using Theorem 2.5 and a similar discussion as [31], we can obtain the following result.

Corollary 2.7 *Let $T = T[A, B]$ be a tree with $a = |A| \leq |B|$ and $\delta(A) = k \geq 2$, and let T_0 be a good odd-ballooning of T such that $\delta_1(A) = k_1 < k$. If n is sufficiently large, then $\text{EX}(n, T_0) = \{G_{n,a,k}\}$ for $(k, k_1) \neq (4, 3)$, and $\text{EX}(n, T_0) = \{G_{n,a,4}, G'_{n,a,4}\}$ for $(k, k_1) = (4, 3)$.*

For $2 \leq s \leq t$, let H be the graph obtained from $K_{t-1,t-1}$ by deleting a matching of $t-2$ edges. Let $G_{s,t}$ be the graph obtained from $K_{s-1} \otimes T(n-s+1, 2)$ by embedding H into one class of $T(n-s+1, 2)$. Let $G'_{3,3}$ be the graph obtained from $K_2 \otimes T(n-2, 2)$ by embedding a triangle into one class of $T(n-2, 2)$. Using Theorem 1.1, Peng and Xia [18] obtained the following result.

Theorem 2.8 ([18]) *Let $F_{s,t}$ be the odd-ballooning of $K_{s,t}$ with $t \geq s \geq 2$, where $s+t \geq 6$ and each odd cycle has length at least 5. Then for sufficiently large n ,*

$$\text{ex}(n, F_{s,t}) = \lceil \frac{n-s+1}{2} \rceil \lfloor \frac{n-s+1}{2} \rfloor + (s-1)(n-s+1) + \binom{s-1}{2} + t^2 - 3t + 3.$$

Moreover, $\text{EX}(n, F_{s,t}) = \{G_{s,t}\}$ for $t \geq 4$, and $\text{EX}(n, F_{3,3}) \supseteq \{G_{3,3}, G'_{3,3}\}$ for $t = 3$.

Let $G''_{3,3}$ be the graph obtained from $\overline{K_2} \otimes T(n-2, 2)$ by embedding a C_4 into one class of $T(n-2, 2)$. Note that $F_{3,3}$ contains a $F_{1,3}$ after deleting any two non-adjacent vertices. We see that $G''_{3,3}$ is $F_{3,3}$ -free (using [14] or [29]). (If we also consider the case that w_1 and w_2 is not adjacent in Claim 12 of [18], we can find $G''_{3,3}$.) Using Theorem 2.5 and a similar discussion as [18], we can obtain the following result.

Corollary 2.9 *Let $F_{3,3}$ be the odd-ballooning of $K_{3,3}$, such that each odd cycle has length at least 5. Then for sufficiently large n , $\text{EX}(n, F_{3,3}) = \{G_{3,3}, G'_{3,3}, G''_{3,3}\}$.*

3 Preliminaries

To prove the main results of this paper, we first include several lemmas. The following result is the classical stability theorem proved by Erdős, Stone and Simonovits [9, 10]:

Theorem 3.1 ([9, 10]) *Let $\mathcal{F}$ be a finite family of graphs with $\min_{F \in \mathcal{F}} \chi(F) = r + 1 \geq 3$. For every $\epsilon > 0$, there exist $\delta > 0$ and n_0 such that, if G is an $\mathcal{F}$ -free graph of order $n \geq n_0$ with $e(G) > (\frac{r-1}{2r} - \delta)n^2$, then G can be obtained from $T(n, r)$ by adding and deleting at most ϵn^2 edges.*

The following well-known lemma is proved by Erdős [6] and Simonovits [19].

Lemma 3.2 ([6, 19]) *Let $\mathcal{F}$ be a finite family of graphs with $\min_{F \in \mathcal{F}} \chi(F) = r + 1 \geq 3$. For any given $\theta > 0$, if n is sufficiently large, then $\delta(G) > (1 - \frac{1}{r} - \theta)n$ for any $G \in \text{EX}(n, \mathcal{F})$.*

As one of the earliest extremal results in graph theory, Erdős and Gallai [5] proved the following theorem.

Lemma 3.3 ([5]) *For any $\ell \geq 2$ and $n \geq \ell$, let G be a P_ℓ -free graph of order n . Then $e(G) \leq \frac{\ell-2}{2}n$ with equality if and only if $n = (\ell-1)t$ and $G = \cup_{1 \leq i \leq t} K_{\ell-1}$.*

Faudree and Schelp [11] improved this result as follows:

Lemma 3.4 ([11]) *Assume that $\ell \geq 2$ and $n = (\ell-1)t + s$, where $0 \leq s < \ell-1$. Let G be a P_ℓ -free graph of order n . Then $e(G) \leq t \binom{\ell-1}{2} + \binom{s}{2}$ with equality if and only if either $G = (\cup_{1 \leq i \leq t} K_{\ell-1}) \cup K_s$, or $G = (\cup_{1 \leq i \leq t-t_0} K_{\ell-1}) \cup (K_{\frac{\ell}{2}-1} \vee \overline{K_{(\ell-1)t_0 - \frac{\ell}{2} + s + 1}})$ for some $1 \leq t_0 \leq t$ when ℓ is even and $s = \frac{\ell}{2} \pm 1$.*

Simonovits [19] introduced the progressive induction.

Lemma 3.5 ([19]) *Let $\mathfrak{U} = \cup_{1 \leq n \leq \infty} \mathfrak{U}_n$ be a set of given elements, where $\mathfrak{U}_n$ are disjoint finite subsets of $\mathfrak{U}$. Let B be a condition or property defined on $\mathfrak{U}$ (i.e., the elements of $\mathfrak{U}$ may satisfy or not satisfy B). Let $\phi(a)$ be a non-negative integer-valued function defined on $\mathfrak{U}$, such that*

- (i) *if a satisfies B , then $\phi(a) = 0$;*
- (ii) *There is a M_0 , such that for any $n > M_0$ and $a \in \mathfrak{U}_n$, either a satisfies B , or there exist an n' with $\frac{n}{2} < n' < n$ and an $a' \in \mathfrak{U}_{n'}$ such that $\phi(a) < \phi(a')$.*

Then there exists an N_0 such that any $a \in \mathfrak{U}_n$ with $n > N_0$ satisfies B .

The following result is a consequence of Lemma 3.2.1 in [20]. Furthermore, Simonovits pointed out that the orders of the symmetric subgraphs are bounded (i.e., $O(1)$) at the end of the proof of Lemma 3.2.1 in [20].

Lemma 3.6 ([20]) *Let G be a P_ℓ -free graph of order n . For any sequence of positive integers $\pi = \{m_k\}_{k \geq 1}$, there exist two positive integers $C(\ell, \pi)$ and $D(\ell, \pi)$ such that if $n \geq C(\ell, \pi)$, then G has m_k symmetric subgraphs $Q_1, Q_2, \dots, Q_{m_k}$, where $1 \leq |Q_1| = k \leq D(\ell, \pi)$.*

The following result will be used in our proofs.

Lemma 3.7 *For $\ell \geq 3$, let G be a P_ℓ -free graph. Suppose that $S \subseteq V(G)$ satisfies that $\frac{|S|}{\ell-1}$ is an integer, and $G[S] = \cup_{1 \leq i \leq \tau} Q_i$, where $Q_1, Q_2, \dots, Q_\tau$ are symmetric subgraphs of G with $\tau \geq \ell$. If $e(G[S]) + e_G(S, V(G) - S) \geq \frac{\ell-2}{2}|S|$, then $e(G[S]) + e_G(S, V(G) - S) = \frac{\ell-2}{2}|S|$, and we have the following conclusions.*

- (i) *For odd $\ell \geq 3$, $Q_i = K_{\ell-1}$ for $1 \leq i \leq \tau$ and $e_G(S, V(G) - S) = 0$.*
- (ii) *For even $\ell \geq 4$, either $Q_i = K_{\ell-1}$ for any $1 \leq i \leq \tau$ and $e_G(S, V(G) - S) = 0$, or $Q_i = K_1$ for any $1 \leq i \leq \tau$ and there are $\frac{\ell-2}{2}$ vertices in $V(G) - S$ which are all adjacent to the (unique) vertex in Q_i for all $1 \leq i \leq \tau$.*

Proof: Let $B \subseteq V(G) - S$ be the set of vertices which have at least one neighbor in Q_1 . Let H be the graph obtained from $G[S \cup B]$ by adding some edges inside B such that $H[B]$ is a complete graph. Let $b = |B|$. If $b = 0$, then $e(G[S]) \geq \frac{\ell-2}{2}|S|$. Note that $G[S]$ is P_ℓ -free. By Lemma 3.3, we must have $e(G[S]) = \frac{\ell-2}{2}|S|$ and $Q_i = K_{\ell-1}$ for any $1 \leq i \leq \tau$. Now we assume $b \geq 1$. Clearly, $|H| = |S| + b \geq \ell$ and $e(H) = e(G[S]) + e_G(S, V(G) - S) + \binom{b}{2} \geq \frac{\ell-2}{2}|S| + \binom{b}{2}$.

By Symmetry, each vertex in B has at least one neighbor in Q_i for any $1 \leq i \leq \tau$. Since $b \geq 1$, H is connected. Now we show that H is P_ℓ -free. Suppose not, i.e., there is a P_ℓ in H , say $P = u_1 u_2 \dots u_\ell$. We choose such a P that P contains as small number of edges in $H[B]$ as possible. If P contains no edges inside B , then P is also a path in G , a

contradiction. Hence, we can assume that P contains at least one edge in $H[B]$. Without loss of generality, assume that u_2u_3 is an edge in $H[B]$. Noting $\tau \geq \ell$, we can choose one symmetric subgraph, say Q_τ , such that P contains no vertices in $V(Q_\tau)$. Since $u_2, u_3 \in B$ have neighbors in Q_τ , let $v, w \in V(Q_\tau)$ be neighbors of u_2, u_3 , respectively. Let $P(v, w)$ be a path from v to w in Q_τ . Then $u_1u_2P(v, w)u_3 \cdots u_\ell$ is a path with less number of edges inside B than P . Clearly, its sub-path of order ℓ is a P_ℓ with less number of edges inside B than P . This contradicts the choice of P . Hence, H is P_ℓ -free. Recall that $|H| = |S| + b$ and $e(H) \geq \frac{\ell-2}{2}|S| + \binom{b}{2}$. Since H is connected and $\frac{|S|}{\ell-1}$ is an integer, by Lemma 3.4, we must have that ℓ is even and $H = K_{\frac{\ell-2}{2}} \otimes \overline{K_{|S|+b-\frac{\ell-2}{2}}}$. Note that H will contain no edges after deleting $\frac{\ell-2}{2}$ vertices in the part $V(K_{\frac{\ell-2}{2}})$. Since $\tau \geq \ell$, we must have $|Q_i| = 1$ for any $1 \leq i \leq \tau$. Moreover, $|B| = \frac{\ell-2}{2}$ and each vertex in B is adjacent to the unique vertex in $|Q_i| = 1$ for any $1 \leq i \leq \tau$. This completes the proof. $\square$

4 Proof of Lemma 1.2

The following fact will be used.

Fact 1. Let $A_1, A_2, \dots, A_m$ be $m \geq 2$ finite sets. Then

$$|\cap_{1 \leq i \leq m} A_i| \geq (\sum_{1 \leq i \leq m} |A_i|) - (m-1)|\cup_{1 \leq i \leq m} A_i|.$$

Proof of Lemma 1.2. Recall that $\min_{F \in \mathcal{F}} \chi(F) = r+1 \geq 3$ and $t = \max_{F \in \mathcal{F}} |F|$. There are graphs $F_1, F_2 \in \mathcal{F}$ such that

$$F_1 \subseteq P_t \otimes T(t(r-1), r-1)$$

and

$$F_2 \subseteq tK_{1,t} \otimes tK_{1,t} \prod T(t(r-2), r-2).$$

For sufficiently large n , let $G \in \text{EX}(n, \mathcal{F})$. Let $\theta := \theta(r, t) > 0$ be a small constant (such that all the following inequalities on it are satisfied). By Lemma 3.2, we have that $\delta(G) > (\frac{r-1}{r} - \theta)n$. As is well known, $e(T(n, r)) \geq \frac{r-1}{2r}n^2 - \frac{r}{8}$. Since $T(n, r)$ is $\mathcal{F}$ -free, we have $e(G) \geq e(T(n, r)) \geq \frac{r-1}{2r}n^2 - \frac{r}{8}$. Since G is $\mathcal{F}$ -free and n is large, by Theorem 3.1 (letting $\epsilon = \theta^3$), we see that G can be obtained from $T(n, r)$ by deleting and adding at most $\theta^3 n^2$ edges. It follows that

$$(\frac{r-1}{2r} - \theta^3)n^2 \leq e(G) \leq (\frac{r-1}{2r} + \theta^3)n^2.$$

Let $V(G) = \cup_{1 \leq i \leq r} V_i$ be a partition such that $\sum_{1 \leq i \leq r} e(G[V_i])$ is minimum. Recall that G can be obtained from $T(n, r)$ by deleting at most $\theta^3 n^2$ edges. Thus,

$$\sum_{1 \leq i \leq r} e(G[V_i]) \leq \theta^3 n^2.$$

Claim 1. $||V_i| - \frac{n}{r}| \leq \theta n$ for any $1 \leq i \leq r$.

Proof of Claim 1. Let $a = \max_{1 \leq i \leq r} ||V_i| - \frac{n}{r}|$. Without loss of generality, assume that $a = ||V_1| - \frac{n}{r}|$. Using the Cauchy-Schwarz inequality, we see

$$2 \sum_{2 \leq i < j \leq r} |V_i||V_j| = \left(\sum_{2 \leq i \leq r} |V_i| \right)^2 - \sum_{2 \leq i \leq r} |V_i|^2 \leq \frac{r-2}{r-1} \left(\sum_{2 \leq i \leq r} |V_i| \right)^2 = \frac{r-2}{r-1} (n - |V_1|)^2.$$

Then, noting $a = ||V_1| - \frac{n}{r}|$,

$$\begin{aligned} e(G) &\leq \left(\sum_{1 \leq i < j \leq r} |V_i||V_j| \right) + \left(\sum_{1 \leq i \leq r} e(G[V_i]) \right) \\ &\leq |V_1|(n - |V_1|) + \left(\sum_{2 \leq i < j \leq r} |V_i||V_j| \right) + \theta^3 n^2 \\ &\leq |V_1|(n - |V_1|) + \frac{r-2}{2(r-1)} (n - |V_1|)^2 + \theta^3 n^2 \\ &= \frac{r-1}{2r} n^2 - \frac{r}{2(r-1)} a^2 + \theta^3 n^2. \end{aligned}$$

Recall that $e(G) \geq (\frac{r-1}{2r} - \theta^3) n^2$. Thus, $(\frac{r-1}{2r} - \theta^3) n^2 \leq \frac{r-1}{2r} n^2 - \frac{r}{2(r-1)} a^2 + \theta^3 n^2$, implying that $a \leq \sqrt{\frac{4(r-1)}{r} \theta^3 n^2} \leq \theta n$. This finishes the proof. $\square$

For any $1 \leq i \leq r$, let $W_i = \{v \in V_i \mid d_{V_i}(v) \geq \theta n\}$, and let $\bar{V}_i = V_i - W_i$. Set $W = \cup_{1 \leq i \leq r} W_i$.

Claim 2. $|W| \leq 2\theta^2 n$.

Proof of Claim 2. Since $\sum_{1 \leq i \leq r} e(G[V_i]) \leq \theta^3 n^2$, and

$$\sum_{1 \leq i \leq r} e(G[V_i]) = \sum_{1 \leq i \leq r} \frac{1}{2} \sum_{v \in V_i} d_{V_i}(v) \geq \sum_{1 \leq i \leq r} \frac{1}{2} \sum_{v \in W_i} d_{V_i}(v) \geq \frac{1}{2} \sum_{1 \leq i \leq r} \theta n |W_i| = \frac{1}{2} \theta n |W|,$$

we have $\frac{1}{2} \theta n |W| \leq \theta^3 n^2$, implying that $|W| \leq 2\theta^2 n$. $\square$

Claim 3. Let $1 \leq \ell \leq r$ be fixed. Assume that $u_1, u_2, \dots, u_{2rt^2} \in \cup_{1 \leq i \neq \ell \leq r} \bar{V}_i$. For any constant $D_0 > t$, there are D_0 vertices in $\bar{V}_\ell$ which are the common neighbors of $u_1, u_2, \dots, u_{2rt^2}$ in G .

Proof of Claim 3. For any $1 \leq i \leq 2rt^2$, assume that $u_i \in \overline{V}_{j_i}$, where $j_i \neq \ell$. Then $d_G(u_i) \geq (\frac{r-1}{r} - \theta)n$ and $d_{V_{j_i}}(u_i) \leq \theta n$ as $u_i \notin W$. Recall that $|V_s| \leq \frac{n}{r} + \theta n$ for any $1 \leq s \leq r$. Hence

$$d_{V_\ell}(u_i) = d_G(u_i) - d_{V_{j_i}}(u_i) - \sum_{1 \leq s \neq \ell, j_i \leq r} d_{V_s}(u_i) \geq (\frac{1}{r} - r\theta)n.$$

Then, using Fact 1, we have

$$|\cap_{1 \leq i \leq 2rt^2} N_{V_\ell}(u_i)| \geq (\sum_{1 \leq i \leq 2rt^2} |N_{V_\ell}(u_i)|) - (2rt^2 - 1)|V_\ell| \geq (\frac{1}{r} - 2r(r+1)t^2\theta)n.$$

It follows that

$$|\cap_{1 \leq i \leq 2rt^2} N_{\overline{V}_\ell}(u_i)| \geq (\frac{1}{r} - 2r(r+1)t^2\theta)n - |W| \geq (\frac{1}{r} - 2r(r+2)t^2\theta)n > D_0.$$

Thus, there are D_0 vertices in $\overline{V}_\ell$ which are all adjacent to the vertices $u_1, u_2, \dots, u_{2rt^2}$. $\square$

Since G is $P_t \otimes T((r-1)t, r-1)$ -free, by Claim 3, we see $G[\overline{V}_i]$ is P_t -free for any $1 \leq i \leq r$. The following claim can be obtained using the method in [19]. For completeness, we give a proof here.

Claim 4. $|W| \leq M$, where $M = \lceil 4^{(r-1)t}(\frac{2}{\theta})^t rt \rceil$ is a constant.

Proof of Claim 4. Recall that $|W| \leq 2\theta^2 n$ by Claim 2. For any fixed $1 \leq i \leq r$ and $1 \leq j \neq i \leq r$, we have $d_{V_i}(v) \leq d_{V_j}(v)$ for any $v \in V_i$, since $\sum_{1 \leq \ell \leq r} e(G[V_i])$ is minimum (otherwise, we can obtain a contradiction by moving v from V_i to V_j). Therefore, $d_{V_j}(v) \geq \frac{1}{2}(d_{V_i}(v) + d_{V_j}(v))$ for any $v \in V_i$. Recall that $|V_s| \leq (\frac{1}{r} + \theta)n$ for any $1 \leq s \leq r$. If $v \in W_i$, then

$$d_{V_j}(v) \geq \frac{1}{2}(d_G(v) - \sum_{1 \leq s \neq i, j \leq r} |V_s|) \geq \frac{1}{2}(\frac{r-1}{r}n - \theta n - (r-2)(\frac{1}{r} + \theta)n) \geq (\frac{1}{2r} - \frac{r-1}{2}\theta)n.$$

It follows that

$$d_{\overline{V}_j}(v) \geq d_{V_j}(v) - |W| \geq (\frac{1}{2r} - \frac{r-1}{2}\theta)n - 2\theta^2 n \geq \frac{n}{2r} - \frac{r}{2}\theta n.$$

If $v \in \overline{V}_i$, then $d_{V_i}(v) \leq \theta n$. Thus

$$d_{V_j}(v) \geq d_G(v) - d_{V_i}(v) - \sum_{1 \leq s \neq i, j \leq r} |V_s| \geq (\frac{r-1}{r} - \theta)n - \theta n - (r-2)(\frac{1}{r} + \theta)n \geq (\frac{1}{r} - r\theta)n.$$

It follows that

$$d_{\overline{V}_j}(v) \geq d_{V_j}(v) - |W| \geq (\frac{1}{r} - r\theta)n - 2\theta^2 n \geq \frac{n}{r} - (r+1)\theta n.$$

In the following, we first show that $|W_1| \leq 4^{(r-1)t}(\frac{2}{\theta})^t t$.

For any $v \in W_1$, we have that $d_{\bar{V}_1}(v) \geq d_{V_1}(v) - |W_1| \geq \theta n - 2\theta^2 n \geq \frac{1}{2}\theta n$. For $j \neq 1$, we have $d_{\bar{V}_j}(v) \geq \frac{n}{2r} - \frac{r}{2}\theta n$ by the above discussion. Let

$$Y_1 = \{(w, Z) \mid w \in W_1, Z \subseteq \bar{V}_1, |Z| = t\},$$

where w is adjacent to all the vertices in Z . Since $d_{\bar{V}_1}(w) \geq \frac{1}{2}\theta n$ for any $w \in W_1$, we have $|Y_1| \geq \binom{\frac{1}{2}\theta n}{t}|W_1|$. On the other hand, $\bar{V}_1$ has at most $\binom{(\frac{1}{r}+\theta)n}{t}$ subsets Z of size t . Hence, there is a subset $Z_1 \subseteq \bar{V}_1$ of size t such that the vertices in Z_1 have at least $\frac{\binom{\frac{1}{2}\theta n}{t}|W_1|}{\binom{(\frac{1}{r}+\theta)n}{t}} \geq (\frac{\theta}{2})^t |W_1|$ common neighbors in W_1 . Let $W^1 \subseteq W_1$ be the set of common neighbors of the vertices in Z_1 . Then $|W^1| \geq (\frac{\theta}{2})^t |W_1|$.

Suppose that for some $1 \leq s \leq r-1$, we have obtained several sets $Z_i \subseteq \bar{V}_i$ of size t for any $1 \leq i \leq s$ and $W^s \subseteq W_1$, such that any two vertices from distinct parts of $W^s, Z_1, \dots, Z_s$ are adjacent in G and $|W^s| \geq (\frac{1}{4})^{(s-1)t}(\frac{\theta}{2})^t |W_1|$. Let $V'_{s+1} \subseteq \bar{V}_{s+1}$ be the set of common neighbors of the vertices in $\cup_{1 \leq i \leq s} Z_i$. Since $(\frac{n}{r} - 2\theta)n \leq |\bar{V}_{s+1}| \leq (\frac{n}{r} + 2\theta)n$ and $d_{\bar{V}_{s+1}}(u) \geq \frac{n}{r} - (r+1)\theta n$ for any $u \in \cup_{1 \leq i \leq s} Z_i$ by the above discussion, we have that $d_{\bar{V}_{s+1}}(u) \geq |\bar{V}_{s+1}| - (r+3)\theta n$ for any $u \in \cup_{1 \leq i \leq s} Z_i$. Using Fact 1, we have

$$|V'_{s+1}| = \left| \bigcap_{u \in \cup_{1 \leq i \leq s} Z_i} N_{\bar{V}_{s+1}}(u) \right| \geq |\bar{V}_{s+1}| - st(r+3)\theta n \geq |\bar{V}_{s+1}| - rt(r+3)\theta n.$$

For any $w \in W_1$, note that $d_{\bar{V}_{s+1}}(w) \geq \frac{n}{2r} - \frac{r}{2}\theta n$ as above. Then $d_{V'_{s+1}}(w) \geq \frac{n}{2r} - rt(r+4)\theta n$.

Note that $|V'_{s+1}| \leq (\frac{1}{r} + \theta)n$. Let

$$Y_{s+1} = \{(w, Z) \mid w \in W^s, Z \subseteq V'_{s+1}, |Z| = t\},$$

where w is adjacent to all the vertices in Z . Since $d_{V'_{s+1}}(w) \geq \frac{n}{2r} - rt(r+4)\theta n \geq \frac{n}{3r}$ for any $w \in W^s$, we have $|Y_{s+1}| \geq |W^s| \binom{\frac{n}{3r}}{t}$. On the other hand, V'_{s+1} has at most $\binom{(\frac{1}{r}+\theta)n}{t}$ subsets Z of size t . Hence, there is a subset $Z_{s+1} \subseteq V'_{s+1}$ of size t such that the vertices in Z_{s+1} have at least $\frac{|W^s| \binom{\frac{n}{3r}}{t}}{\binom{(\frac{1}{r}+\theta)n}{t}} \geq (\frac{1}{4})^t |W^s|$ common neighbors in W^s . Let $W^{s+1} \subseteq W^s$ be the set of common neighbors of the vertices in Z_{s+1} . Then

$$|W^{s+1}| \geq (\frac{1}{4})^t |W^s| \geq (\frac{1}{4})^{st} (\frac{\theta}{2})^t |W_1|.$$

Therefore, by induction on s , we can obtain several sets $Z_i \subseteq \bar{V}_i$ of size t for any $1 \leq i \leq r$ and $W^r \subseteq W_1$, such that any two vertices from distinct parts of $W^r, Z_1, \dots, Z_r$ are adjacent in G and $|W^r| \geq (\frac{1}{4})^{(r-1)t}(\frac{\theta}{2})^t |W_1|$. So, if $|W_1| \geq 4^{(r-1)t}(\frac{2}{\theta})^t t$, then $|W^r| \geq t$. It follows that the subgraph induced by $W^r, Z_1, \dots, Z_r$ contains a complete $(r+1)$ -partite graph of

which each part has size at least t . This contradicts the fact that G is $\mathcal{F}$ -free. Thus $|W_1| < 4^{(r-1)t} \left(\frac{2}{\theta}\right)^t t$. Similarly, we can show that $|W_i| < 4^{(r-1)t} \left(\frac{2}{\theta}\right)^t t$ for any $2 \leq i \leq r$. Hence $|W| < 4^{(r-1)t} \left(\frac{2}{\theta}\right)^t r t$. This finishes the proof. $\square$

For any fixed $1 \leq i \leq r$, let Q be an induced subgraph of $G[\bar{V}_i]$ (or a subset of $\bar{V}_i$). Define

$$\alpha(Q) = E(Q) \cup \{uv \in E(G) \mid u \in V(Q), v \in (V_i - V(Q)) \cup W\},$$

and

$$\beta(Q) = \{uv \notin E(G) \mid u \in V(Q), v \in \cup_{1 \leq j \neq i \leq r} \bar{V}_j\}.$$

Lemma 4.1 *For any fixed $1 \leq i_0 \leq r$, if $Q_1, Q_2, \dots, Q_t$ are symmetric subgraphs of $G[\bar{V}_{i_0}]$, then $|\alpha(Q_i)| \leq (M + 2t)|Q_1|$ and $|\beta(Q_1)| \leq (M + 2t)|Q_1|$.*

Proof: For convenience, assume $i_0 = 1$. We first show that there are at most t vertices in $\bar{V}_1 - \cup_{1 \leq i \leq t} V(Q_i)$ which have neighbors in $V(Q_1)$. If not, there are t vertices $u_1, u_2, \dots, u_t$ in $\bar{V}_1 - \cup_{1 \leq i \leq t} V(Q_i)$, such that $u_i v_i$ is an edge for any $1 \leq i \leq t$, where $v_1, v_2, \dots, v_t \in V(Q_1)$ (may be repeated). Since $Q_1, Q_2, \dots, Q_t$ are symmetric subgraphs in $G[\bar{V}_1]$, for $2 \leq j \leq t$ let $v_1^j, v_2^j, \dots, v_t^j \in V(Q_j)$ be the symmetric vertices of $v_1, v_2, \dots, v_t$ in $V(Q_j)$, respectively. Then u_i is adjacent to v_i^j for any $1 \leq i, j \leq t$. Note that Q_j is connected for any $1 \leq j \leq t$. Let $P_{v_j^j v_{j+1}^j}$ be a path connecting v_j^j to v_{j+1}^j in Q_j . Then

$$u_1 P_{v_1^1 v_2^1} u_2 P_{v_2^2 v_3^2} u_3 \cdots u_{t-1} P_{v_{t-1}^{t-1} v_t^{t-1}} u_t$$

is a path of order at least $2t - 1$ in $G[\bar{V}_1]$. Now we select its a sub-path of order t , say $w_1, w_2, \dots, w_t$. By Claim 3, these t vertices have t common neighbors in $\bar{V}_2$, say $w_1^2, w_2^2, \dots, w_t^2$. By Claim 3, these $2t$ vertices have t common neighbors in $\bar{V}_3$. Repeat this process. Eventually, we can obtain a subgraph in G , which contains $P_t \otimes T(t(r-1), r-1)$ as a subgraph, a contradiction. Hence there are at most t vertices in $\bar{V}_1 - \cup_{1 \leq i \leq t} V(Q_i)$ which have neighbors in $V(Q_1)$. Moreover, using a similar discussion, we see that Q_i is $\{P_t\}$ -free for any $1 \leq i \leq t$. This implies that $e(Q_1) \leq t|Q_1|$ by Lemma 3.3. Hence, $|\alpha(Q_1)| \leq (M + 2t)|Q_1|$.

Suppose $|\beta(Q_1)| > (M + 2t)|Q_1|$. Let G' be the graph obtained from G by deleting all the edges in $\alpha(Q_1)$, and adding all the non-edges in $\beta(Q_1)$. Then $F \subseteq G'$ for some $F \in \mathcal{F}$, since $e(G') > e(G)$ and $G \in \text{EX}(n, \mathcal{F})$. Let U be the set of vertices of F which are adjacent to vertices in $V(F) \cap V(Q_1)$ in G' . Then $U \subseteq (\cup_{2 \leq i \leq r} \bar{V}_i)$. By Claim 3, in G we can choose t common neighbors in $\bar{V}_1 - V(F)$ of the vertices in U , say $y_1, y_2, \dots, y_t$. Let F' be the graph of G induced by $(V(F) - V(F) \cap V(Q_1)) \cup \{y_1, y_2, \dots, y_t\}$. Clearly, $F \subseteq F'$, a contradiction as G is F -free. Hence, $|\beta(Q_1)| \leq (M + 2t)|Q_1|$. For any Q_i with $2 \leq i \leq t$, we can use a similar discussion. This finishes the proof. $\square$

The idea of Lemma 4.2 should be attributed to Simonovits [20].

Lemma 4.2 *Let $1 \leq i_0 \leq r$ be fixed. Assume that $Q_1, Q_2, \dots, Q_\tau$ are symmetric subgraphs of $G[V_{i_0} \cup (W \cup B)]$, where $B \subseteq V(G) - (V_{i_0} \cup W)$ and $V(Q_i) \subseteq \bar{V}_{i_0}$ for any $1 \leq i \leq \tau$. If $Q_1, Q_2, \dots, Q_\tau$ are not symmetric in G , then there exists one vertex $v \notin V_{i_0} \cup (W \cup B)$, such that at least $\frac{1}{2^{|Q_1|+1}t}\tau$ ones among the τ subgraphs, say $Q_1, Q_2, \dots, Q_{\frac{1}{2^{|Q_1|+1}t}\tau}$, are also symmetric in $G[V_{i_0} \cup (W \cup B \cup \{v\})]$, and v has at least one non-neighbor in Q_i for any $1 \leq i \leq \frac{1}{2^{|Q_1|+1}t}\tau$.*

Proof: We can assume that $\frac{1}{2^{|Q_1|+1}t}\tau \geq 1$. Otherwise, there is nothing to prove. Then $\tau \geq 2^{|Q_1|+1}t$. For convenience, assume $i_0 = 1$. Let $Z = V(G) - (V_1 \cup (W \cup B))$. If all vertices of Q_i are adjacent to all the vertices in Z for any $1 \leq i \leq \tau$, then $Q_1, Q_2, \dots, Q_\tau$ are symmetric in G , too. This contradicts the assumption. Thus, without loss of generality, we can assume that $w_1 \in Z$ has a non-neighbor $u_1 \in V(Q_1)$. Let G' be the graph obtained from G by adding the edge $u_1 w_1$. Since $G \in \text{EX}(n, \mathcal{F})$ and $e(G') > e(G)$, we see that G' contains a subgraph $F \in \mathcal{F}$. Clearly, F contains the edge $u_1 w_1$.

Let $v_1, \dots, v_j$ be all the vertices in Z which are contained in F , where $1 \leq j \leq t$. Now we show that there are at most t ones among $Q_1, Q_2, \dots, Q_\tau$, whose vertices are all adjacent to all the vertices $v_1, \dots, v_j$. Suppose not. Noting $|F| \leq t$, we can find a Q_ℓ with $1 \leq \ell \leq \tau$ and $V(Q_\ell) \cap V(F) = \emptyset$, such that all vertices of Q_ℓ are adjacent to all the vertices $v_1, \dots, v_j$. Recall that $Q_1, Q_2, \dots, Q_\tau$ are symmetric in $G[V_1 \cup (W \cup B)]$. We see that all the neighbors in F of the vertices in $F \cap V(Q_1)$ are adjacent to the corresponding vertices in $V(Q_\ell)$ in G . Let F' be the subgraph of G induced by $(V(F) - V(Q_1)) \cup V(Q_\ell)$. Clearly, $F \subseteq F'$. Thus G contains a copy of F , a contradiction. Therefore, there are at most t ones among $Q_1, Q_2, \dots, Q_\tau$, whose vertices are all adjacent to all the vertices $v_1, \dots, v_j$. Hence, there are at least $\tau - t$ ones among $Q_1, Q_2, \dots, Q_\tau$, which have a non-neighbor in $\{v_1, \dots, v_j\}$. So, we can find a vertex among $v_1, \dots, v_j$, say v , such that v has a non-neighbor in at least $\frac{\tau-t}{j} \geq \frac{\tau-t}{t} \geq \frac{1}{2t}\tau$ ones among $Q_1, Q_2, \dots, Q_\tau$, say $Q_1, Q_2, \dots, Q_{\frac{1}{2t}\tau}$. Clearly, v is connecting in the same way to at least $\frac{1}{2t} \frac{1}{2^{|Q_1|}}\tau = \frac{1}{2^{|Q_1|+1}t}\tau$ ones among these $\frac{1}{2t}\tau$ subgraphs, say $Q_1, Q_2, \dots, Q_{\frac{1}{2^{|Q_1|+1}t}\tau}$. Clearly, $Q_1, Q_2, \dots, Q_{\frac{1}{2^{|Q_1|+1}t}\tau}$ are also symmetric in $G[V_{i_0} \cup (W \cup B \cup \{v\})]$. This finishes the proof. $\square$

Lemma 4.3 *Let $1 \leq i_0 \leq r$ be fixed. Assume that $Q_1, Q_2, \dots, Q_\tau$ are symmetric subgraphs of $G[V_{i_0} \cup W]$, where $V(Q_i) \subseteq \bar{V}_{i_0}$ for any $1 \leq i \leq \tau$. Then there are at least $(\frac{1}{2^{|Q_1|+1}t})^{(M+2t)|Q_1|} \cdot \tau$ ones among $Q_1, Q_2, \dots, Q_\tau$ are symmetric in G , too.*

Proof: We can assume that $(\frac{1}{2^{|Q_1|+1}t})^{(M+2t)|Q_1|}\tau \geq 1$. Otherwise, there is nothing to prove. Then $\tau \geq (2^{|Q_1|+1}t)^{(M+2t)|Q_1|}$. For convenience, assume $i_0 = 1$. By Lemma 4.1, $|\beta(Q_i)| \leq (M+2t)|Q_1|$ for any $1 \leq i \leq \tau$.

If $Q_1, Q_2, \dots, Q_\tau$ are symmetric in G , then the lemma holds. So, we can assume that $Q_1, Q_2, \dots, Q_\tau$ are not symmetric in G . By Lemma 4.2, there exists a vertex $v_1 \notin V_1 \cup W$,

such that at least $\frac{1}{2^{|Q_1|+1}t}\tau$ ones among the τ subgraphs, say $Q_1, Q_2, \dots, Q_{\frac{1}{2^{|Q_1|+1}t}\tau}$, which are symmetric in $G[V_1 \cup (W \cup \{v_1\})]$. Moreover, v_1 has a non-neighbor in $V(Q_i)$ for any $1 \leq i \leq \frac{1}{2^{|Q_1|+1}t}\tau$.

If $Q_1, Q_2, \dots, Q_{\frac{1}{2^{|Q_1|+1}t}\tau}$ are symmetric in G , then the lemma holds. So, we can assume that $Q_1, Q_2, \dots, Q_{\frac{1}{2^{|Q_1|+1}t}\tau}$ are not symmetric in G . By Lemma 4.2 again, there exists another vertex $v_2 \notin V_1 \cup (W \cup \{v_1\})$, such that at least $(\frac{1}{2^{|Q_1|+1}t})^2\tau$ ones among the subgraphs, say $Q_1, Q_2, \dots, Q_{(\frac{1}{2^{|Q_1|+1}t})^2\tau}$, which are symmetric in $G[V_1 \cup (W \cup \{v_1, v_2\})]$. Moreover, v_2 has a non-neighbor in $V(Q_i)$ for any $1 \leq i \leq (\frac{1}{2^{|Q_1|+1}t})^2\tau$.

Repeat the above process. We can obtain vertices $v_1, v_2, \dots, v_\ell \notin V_1 \cup W$, such that $Q_1, Q_2, \dots, Q_{(\frac{1}{2^{|Q_1|+1}t})^\ell \tau}$ are symmetric in $G[V_1 \cup (W \cup \{v_1, v_2, \dots, v_\ell\})]$, and each vertex in $\{v_1, v_2, \dots, v_\ell\}$ has a non-neighbor in $V(Q_i)$ for any $1 \leq i \leq (\frac{1}{2^{|Q_1|+1}t})^\ell \cdot \tau$. Since $|\beta(Q_i)| \leq (M+2t)|Q_1|$ for any $1 \leq i \leq \tau$, we see that $\ell \leq (M+2t)|Q_1|$. Since the process must stop at step $\ell \leq (M+2t)|Q_1|$, we must have that $Q_1, Q_2, \dots, Q_{(\frac{1}{2^{|Q_1|+1}t})^\ell \tau}$ are symmetric in G , too. The proof is completed as

$$\left(\frac{1}{2^{|Q_1|+1}t}\right)^\ell \cdot \tau \geq \left(\frac{1}{2^{|Q_1|+1}t}\right)^{(M+2t)|Q_1|} \cdot \tau.$$

□

Now we come back to the proof Lemma 1.2. Let $f(t) = \sum_{0 \leq i \leq t-1} t^i$. Clearly, a connected graph Q must contain $K_{1,t}$ or P_t as a subgraph if $|Q| \geq f(t)$. Set

$$N = 1 \times 2 \times \dots \times f(t) \times N_0.$$

Let $\pi = \{m_k\}_{k \geq 1}$, where $m_k = 2^{(f(t)(M+2t)r+M)k} \cdot (2^{k+1}t)^{(M+2t)k} \cdot N$ for any $k \geq 1$. Set $D = D(t, \pi) + f(t)$. Let $\pi' = \{m'_k\}_{k \geq 1}$, where $m'_k = 2^{(D(M+2t)r+M)k} \cdot (2^{k+1}t)^{(M+2t)k} \cdot D \cdot N$ for any $k \geq 1$. By Corollary 3.6, if a P_t -free graph H has order at least $C(t, \pi)$, then H has $2^{(f(t)(M+2t)r+M)k} \cdot (2^{k+1}t)^{(M+2t)k} \cdot N$ symmetric subgraphs of order k , where $1 \leq k \leq D$. If a P_t -free graph H has order at least $C(t, \pi')$, then H has $2^{(D(M+2t)r+M)k} \cdot (2^{k+1}t)^{(M+2t)k} \cdot D \cdot N$ symmetric subgraphs of order k , where $1 \leq k \leq D(t, \pi')$. Set $C = \max\{C(t, \pi), C(t, \pi')\}$. In our proof, we can assume that $n \gg C$.

Case 1. For some $1 \leq i_0 \leq r$, there are N symmetric subgraphs $Q_1, Q_2, \dots, Q_N$ of G , such that $\cup_{1 \leq j \leq N} V(Q_j) \subseteq \bar{V}_{i_0}$ and $f(t) \leq |Q_1| \leq D$.

Recall that $Q_1, Q_2, \dots, Q_N$ are symmetric in G with $f(t) \leq |Q_1| \leq D$. Without loss of generality, assume that $i_0 = 1$. That is, $V(Q_i) \subseteq \bar{V}_1$ for any $1 \leq i \leq N$. By Corollary 4.1, $|\beta(Q_i)| \leq (M+2t)|Q_i| \leq (M+2t)D$ for any $1 \leq i \leq N$. Let $\bar{N}(Q_1)$ be the set of vertices in $\bar{V}_2$ which are not adjacent to at least one vertex in $V(Q_1)$. Then $|\bar{N}(Q_1)| \leq |\beta(Q_1)| \leq (M+2t)D$. Let $\bar{V}'_2 = \bar{V}_2 - \bar{N}(Q_1)$. Then by symmetry, all the vertices in $\cup_{1 \leq i \leq N} V(Q_i)$ are adjacent to all the vertices in $\bar{V}'_2$.

Since $|\bar{V}'_2| \geq (\frac{1}{r} - \theta)n - M - (M+2t)D > C$, by Corollary 3.6, $G[\bar{V}'_2]$ has $2^{(D(M+2t)r+M)k} \cdot (2^{k+1}t)^{(M+2t)k} \cdot D \cdot N$ symmetric subgraphs $Q_1^2, Q_2^2, Q_3^2, \dots$, where $|Q_1^2| = k \leq D(t, \pi')$. Set $\tau = 2^{(D(M+2t)r+M)|Q_1^2|} \cdot (2^{|Q_1^2|+1}t)^{(M+2t)|Q_1^2|} \cdot D \cdot N$. Now we prove that $|Q_1^2| < f(t)$. Suppose not, i.e., $|Q_1^2| \geq f(t)$. Then Q_j^2 contains a $K_{1,t}$ for any $1 \leq j \leq t$, since Q_j^2 is P_t -free. Recall that $|Q_j| \geq f(t)$. thus, Q_j contains a $K_{1,t}$ for any $1 \leq j \leq t$. Hence, we obtain a subgraph $A_1 = tK_{1,t}$ in $G[\bar{V}_1]$ and a subgraph $A_2 = tK_{1,t}$ in $G[\bar{V}_2]$. Moreover, the vertices of A_1 are all adjacent to all the vertices of A_2 . By Claim 3, the vertices of A_1, A_2 have t common neighbors in $\bar{V}_3$, say $u_1^3, u_2^3, \dots, u_t^3$. By Claim 3 again, the vertices of $A_1, A_2, \{u_1^3, u_2^3, \dots, u_t^3\}$ have t common neighbors in $\bar{V}_4$, say $u_1^4, u_2^4, \dots, u_t^4$. Repeat the above process. We can obtain a subgraph $tK_{1,t} \otimes tK_{1,t} \otimes T(t(r-2), r-2)$ in G . Hence $|Q_1^2| \leq f(t)$. For any $1 \leq j \leq \tau$, there are at most $2^{|Q_1^2|((M+2t)D+M)}$ ways connecting the vertices of Q_j^2 to the vertices of $W \cup \bar{N}(Q_1)$. Thus, among Q_j^2 with $1 \leq j \leq \tau$, there are at least $2^{-|Q_1^2|((M+2t)D+M)} \cdot \tau$ ones which are also symmetric in $G[V_2 \cup W]$. Denote these ones by $Q_1^2, Q_2^2, \dots, Q_{2^{-|Q_1^2|((M+2t)D+M)} \cdot \tau}^2$ without loss of generality. By Lemma 4.3, among Q_j^2 with $1 \leq j \leq 2^{-|Q_1^2|((M+2t)D+M)} \cdot \tau$, there are at least $(\frac{1}{2^{|Q_1^2|+1}t})^{(M+2t)|Q_1^2|} \cdot 2^{-|Q_1^2|((M+2t)D+M)} \cdot \tau \geq D \cdot N$ ones which are also symmetric in G . Note that N is divisible by $|Q_1^2|$ as $|Q_1^2| \leq f(t)$. It follows that $\frac{|Q_1| \cdot N}{|Q_1^2|} \leq D \cdot N$ is an integer. Thus, without loss of generality, we can assume that $Q_1^2, Q_2^2, \dots, Q_{\frac{|Q_1| \cdot N}{|Q_1^2|}}^2$ are symmetric in G .

Let $\bar{N}(Q_1 \cup Q_1^2)$ be the set of vertices in $\bar{V}_3$ which are not adjacent to at least one vertex in $V(Q_1 \cup Q_1^2)$. Then $|\bar{N}(Q_1 \cup Q_1^2)| \leq |\beta(Q_1 \cup Q_1^2)| \leq 2(M+2t)D$. Let $\bar{V}'_3 = \bar{V}_3 - \bar{N}(Q_1 \cup Q_1^2)$. Then by symmetry, all the vertices in $(\cup_{1 \leq i \leq N} V(Q_i)) \cup (\cup_{1 \leq i \leq \frac{|Q_1| \cdot N}{|Q_1^2|}} V(Q_i^2))$ are adjacent to all the vertices in $\bar{V}'_3$. Since $|\bar{V}'_3| \geq (\frac{1}{r} - \theta)n - M - 2(M+2t)D > C$, by Corollary 3.6, $G[\bar{V}'_3]$ has $2^{(D(M+2t)r+M)|Q_1^3|} \cdot (2^{|Q_1^3|+1}t)^{(M+2t)|Q_1^3|} \cdot D \cdot N$ symmetric subgraphs $Q_1^3, Q_2^3, Q_3^3, \dots$, where $|Q_1^3| \leq D(t, \pi')$. Set $\tau_1 = 2^{(D(M+2t)r+M)|Q_1^3|} \cdot (2^{|Q_1^3|+1}t)^{(M+2t)|Q_1^3|} \cdot D \cdot N$. Similar to the previous paragraph, we can prove that $|Q_1^3| < f(t)$. Moreover, among Q_j^3 with $1 \leq j \leq \tau_1$, we can assume that $Q_1^3, Q_2^3, \dots, Q_{\frac{|Q_1| \cdot N}{|Q_1^3|}}^3$ are symmetric in G .

Repeat the above process. We can find symmetric subgraphs in G :

$$\begin{aligned}
& Q_1, Q_2, \dots, Q_N, \\
& Q_1^2, Q_2^2, \dots, Q_{\frac{|Q_1| \cdot N}{|Q_1^2|}}^2, \\
& Q_1^3, Q_2^3, \dots, Q_{\frac{|Q_1| \cdot N}{|Q_1^3|}}^3, \\
& \dots \dots \dots \\
& Q_1^r, Q_2^r, \dots, Q_{\frac{|Q_1| \cdot N}{|Q_1^r|}}^r.
\end{aligned}$$

Moreover, all the vertices from distinct groups of symmetric subgraphs are adjacent in G . Each group of symmetric subgraphs has exactly $|Q_1|N$ vertices. Let $|Q_1|N = O(r, t) \cdot N_0$. Clearly, $O(r, t)$ is an integer, and $|Q_1|, |Q_1^j| \leq O(r, t)$ for any $2 \leq j \leq r$, as desired.

Case 2. For any $1 \leq i \leq r$, if $Q_1, Q_2, \dots, Q_N$ are symmetric subgraphs of G such that $\cup_{1 \leq j \leq N} V(Q_j) \subseteq \bar{V}_i$, then $|Q_1| < f(t)$ or $|Q_1| > D$.

Ignore the symbols used in Case 1. Since $|\bar{V}_1| \geq (\frac{1}{r} - \theta)n - M > C$, by Corollary 3.6, $G[\bar{V}_1]$ has $2^{(f(t)(M+2t)r+M)k} \cdot (2^{k+1}t)^{(M+2t)k} \cdot N$ symmetric subgraphs $Q_1^1, Q_2^1, Q_3^1, \dots$, where $|Q_1^1| = k \leq D$. Set $\tau_1 = 2^{(f(t)(M+2t)r+M)|Q_1^1|} \cdot (2^{|Q_1^1|+1}t)^{(M+2t)|Q_1^1|} \cdot N$. For any $1 \leq j \leq \tau_1$, there are at most $2^{|Q_1^1|M}$ ways connecting the vertices of Q_j^1 to the vertices of W . Thus, among Q_j^1 with $1 \leq j \leq \tau_1$, there are at least $2^{-|Q_1^1|M} \cdot \tau_1$ ones which are also symmetric in $G[V_1 \cup W]$. Denote these ones by $Q_1^1, Q_2^1, \dots, Q_{2^{-|Q_1^1|M} \cdot \tau_1}^1$ without loss of generality. By Lemma 4.3, among Q_j^1 with $1 \leq j \leq 2^{-|Q_1^1|M} \cdot \tau_1$, there are at least $(\frac{1}{2^{|Q_1^1|+1}t})^{(M+2t)|Q_1^1|} \cdot 2^{-|Q_1^1|M} \cdot \tau_1 \geq N$ ones which are also symmetric in G . Denote these ones by $Q_1^1, Q_2^1, \dots, Q_N^1$ without loss of generality. Since $|Q_1^1| \leq D$, we must have $|Q_1^1| \leq f(t)$ by the assumption in this case.

Let $\bar{N}(Q_1^1)$ be the set of vertices in $\bar{V}_2$ which are not adjacent to at least one vertex in $V(Q_1^1)$. Then $|\bar{N}(Q_1^1)| \leq |\beta(Q_1^1)| \leq (M+2t)|Q_1^1| \leq f(t)(M+2t)$. Let $\bar{V}_2' = \bar{V}_2 - \bar{N}(Q_1^1)$. Then by symmetry, all the vertices in $\cup_{1 \leq i \leq N} V(Q_i^1)$ are adjacent to all the vertices in $\bar{V}_2'$. Since $|\bar{V}_2'| \geq (\frac{1}{r} - \theta)n - M - f(t)(M+2t) > C$, by Corollary 3.6, $G[\bar{V}_2']$ has $2^{(f(t)(M+2t)r+M)k} \cdot (2^{k+1}t)^{(M+2t)k} \cdot N$ symmetric subgraphs $Q_1^2, Q_2^2, Q_3^2, \dots$, where $|Q_1^2| = k \leq D$. Set $\tau_2 = 2^{(f(t)(M+2t)r+M)|Q_1^2|} \cdot (2^{|Q_1^2|+1}t)^{(M+2t)|Q_1^2|} \cdot N$. For any $1 \leq j \leq \tau_2$, there are at most $2^{|Q_1^2|(f(t)(M+2t)+M)}$ ways connecting the vertices of Q_j^2 to the vertices of $W \cup \bar{N}(Q_1^1)$. Thus, among Q_j^2 with $1 \leq j \leq \tau_2$, there are at least $2^{-|Q_1^2|(f(t)(M+2t)+M)} \cdot \tau_2$ ones which are also symmetric in $G[V_2 \cup W]$. Denote these ones by $Q_1^2, Q_2^2, \dots, Q_{2^{-|Q_1^2|(f(t)(M+2t)+M)} \cdot \tau_2}^2$ without loss of generality. By Lemma 4.3, among Q_j^2 with $1 \leq j \leq 2^{-|Q_1^2|(f(t)(M+2t)+M)} \cdot \tau_2$, there are at least $(\frac{1}{2^{|Q_1^2|+1}t})^{(M+2t)|Q_1^2|} \cdot 2^{-|Q_1^2|(f(t)(M+2t)+M)} \cdot \tau_2 \geq N$ ones which are also symmetric in G . Denote these ones by $Q_1^2, Q_2^2, \dots, Q_N^2$ without loss of generality. Since $|Q_1^2| \leq D$, we must have $|Q_1^2| \leq f(t)$ by the assumption in this case.

Let $\bar{N}(Q_1^1 \cup Q_1^2)$ be the set of vertices in $\bar{V}_3$ which are not adjacent to at least one vertex in $V(Q_1^1 \cup Q_1^2)$. Then $|\bar{N}(Q_1^1 \cup Q_1^2)| \leq |\beta(Q_1^1 \cup Q_1^2)| \leq 2(M+2t)f(t)$. Let $\bar{V}_3' = \bar{V}_3 - \bar{N}(Q_1^1 \cup Q_1^2)$. Then by symmetry, all the vertices in $(\cup_{1 \leq i \leq N} V(Q_i^1)) \cup (\cup_{1 \leq i \leq N} V(Q_i^2))$ are adjacent to all the vertices in $\bar{V}_3'$. Since $|\bar{V}_3'| \geq (\frac{1}{r} - \theta)n - M - 2(M+2t)f(t) > C$, by Corollary 3.6, $G[\bar{V}_3']$ has $2^{(f(t)(M+2t)r+M)|Q_1^3|} \cdot (2^{|Q_1^3|+1}t)^{(M+2t)|Q_1^3|} \cdot N$ symmetric subgraphs $Q_1^3, Q_2^3, Q_3^3, \dots$, where $|Q_1^3| \leq D$. Similar to the previous paragraph, we can find symmetric subgraphs of G : $Q_1^3, Q_2^3, \dots, Q_N^3$, such that $|Q_1^3| \leq f(t)$.

Repeat the above process. We can find symmetric subgraphs in G :

$$Q_1^1, Q_2^1, \dots, Q_N^1,$$

$$Q_1^2, Q_2^2, \dots, Q_N^2,$$

$$\dots\dots\dots$$

$$Q_1^r, Q_2^r, \dots, Q_N^r,$$

where $|Q_1^j| \leq f(t)$ for any $1 \leq j \leq r$. Moreover, all the vertices from distinct groups of symmetric subgraphs are adjacent in G . Without loss of generality, assume that $|Q_1| = \min_{1 \leq j \leq r} |Q_j|$. Note that N is divisible by $|Q_j|$. Thus, $\frac{|Q_1|N}{|Q_j|} \leq N$ is an integer for any $1 \leq j \leq r$. For any $2 \leq j \leq r$, we only need $Q_1^j, Q_2^j, \dots, Q_{\frac{|Q_1|N}{|Q_j|}}^j$. Hence, we can obtain r new groups of symmetric subgraphs of G , in which each group has exactly $|Q_1|N$ vertices. Let $|Q_1|N = O(r, t) \cdot N_0$. Clearly, $O(r, t)$ is an integer, and $|Q_1^j| \leq O(r, t)$ for any $1 \leq j \leq r$, as desired. This completes the proof. $\square$

5 Proof of Theorem 1.3

In this section, we give the proof of Theorem 1.3.

Proof of Theorem 1.3. Recall $q = q(\mathcal{F})$. For large m , Let $\mathcal{G}_m$ be the set of $\mathcal{F}$ -free graphs H such that, H has a partition $V(H) = W \cup (\cup_{1 \leq i \leq r} S_i)$ satisfying:

- (i) $|W| = q - 1$ and $|S_i| = \lfloor \frac{n-q+1}{r} \rfloor$ or $\lceil \frac{n-q+1}{r} \rceil$ for any $1 \leq i \leq r$;
- (ii) for any $1 \leq i \leq r$, there exists $S'_i \subseteq S_i$ such that $|S_i - S'_i| \leq t^2$, and $N_H(v) = V(G) - S_i$ for any $v \in S'_i$.

Note that $\mathcal{G}_m$ is not empty, since $\overline{K_{q-1}} \otimes T(n - q + 1, r) \in \mathcal{G}_m$ by the definition of q .

For the graphs in $\mathcal{G}_m$, we can define an operation $\mathcal{D}$ as follows: Let H be a graph in $\mathcal{G}_m$ with the partition as above. Define $\mathcal{D}(H)$ to be the graph obtained from H by adding a new vertex x_i to S_i for each $1 \leq i \leq r$, such that $N_{\mathcal{D}(H)}(x_i) = V(\mathcal{D}(H)) - (S_i \cup \{x_i\})$. Clearly, $\mathcal{D}(H) \in \mathcal{G}_{m+r}$, and $e(\mathcal{D}(H)) = e(H) + (q - 1) + m(r - 1) + \binom{r}{2}$. The inverse of $\mathcal{D}$ is defined as follows: Let H be a graph in $\mathcal{G}_m$ with the partition as above. Define $\mathcal{D}^{-1}(H)$ to be the graph obtained from H by deleting a vertex in S_i for each $1 \leq i \leq r$. Clearly, $\mathcal{D}^{-1}(H) \in \mathcal{G}_{m-r}$, and $e(\mathcal{D}^{-1}(H)) = e(H) - (q - 1) + (m - r)(r - 1) + \binom{r}{2}$. Note that $\mathcal{D}^{-1}(\mathcal{D}(H)) = \mathcal{D}(\mathcal{D}^{-1}(H)) = H$ for any $H \in \mathcal{G}_m$. For any $\ell \geq 1$, define iteratively $\mathcal{D}^{\ell+1}(H) = \mathcal{D}(\mathcal{D}^\ell(H))$, where $\mathcal{D}^1(H) = \mathcal{D}(H)$.

Claim 1. Let H be an extremal graph with maximum edges in $\mathcal{G}_m$. Then $\mathcal{D}^{\pm 1}(H)$ is an extremal graph with maximum edges in $\mathcal{G}_{m \pm r}$.

Proof of Claim 1. For similarity, we only prove the result for $\mathcal{D}^{-1}(H)$. Suppose that $\mathcal{D}^{-1}(H)$ is not an extremal graph in $\mathcal{G}_{m-r}$. Then there is a graph H' in $\mathcal{G}_{m-r}$ with $e(H') > e(\mathcal{D}^{-1}(H))$. Then

$$\begin{aligned} \mathcal{D}(H') &= e(H') + (q-1) + (m-r)(r-1) + \binom{r}{2} \\ &> e(\mathcal{D}^{-1}(H)) + (q-1) + (m-r)(r-1) + \binom{r}{2} = e(H). \end{aligned}$$

Note that $H, \mathcal{D}(H') \in \mathcal{G}_m$. This contradicts the fact that H is extremal in $\mathcal{G}_m$. Hence, $\mathcal{D}^{-1}(H)$ is extremal in $\mathcal{G}_{m-r}$. This finishes the proof of Claim 1. $\square$

For any constant $\ell \geq 1$ and any extremal graph H in $\mathcal{G}_m$ (with large m), by Claim 1, $\mathcal{D}^{-\ell}(H)$ is extremal in $\mathcal{G}_{m-r\ell}$. We shall prove that any graph in $\text{EX}(n, \mathcal{F})$ with large n is in $\mathcal{G}_n$ using the progressive induction.

For $m \geq 1$, let $\mathfrak{U}_m = \text{EX}(m, \mathcal{F})$. For a graph $G' \in \text{EX}(m, \mathcal{F})$, we say G' has property B , if $G' \in \mathcal{G}_m$. Choose an H_m as an extremal graph in $\mathcal{G}_m$. For any $G_m \in \text{EX}(n, \mathcal{F})$, define $\phi(G_m) = e(G_m) - e(H_m)$. Clearly, $\phi(G_m)$ is a non-negative integer-valued function. If G_m has property B , then $\phi(G_m) = 0$. By Lemma 3.5, it suffices to show that for sufficiently large m , either G_m satisfies property B , or there is a $\frac{m}{2} \leq m' < m$, such that $G_{m'} \in \text{EX}(m', \mathcal{F})$ and $\phi(G_{m'}) > \phi(G_m)$.

Recall that $F_1 \subseteq tK_2 \otimes T(t(r-1), r-1)$, where $F_1 \in \mathcal{F}$. Since $|F_1| \leq t$, we see $F_1 \subseteq P_t \otimes T(t(r-1), r-1)$ and $F_1 \subseteq tK_{1,t} \otimes tK_{1,t} \otimes T(t(r-2), r-2)$. Thus $\mathcal{F}$ satisfies the condition in Lemma 1.2. Since $G_m \in \text{EX}(m, \mathcal{F})$ with large m , there is a constant N (by letting $N_0 = 2t$ and $N = O(r, t) \cdot N_0$ in Lemma 1.2), such that G_m has an induced subgraph $\otimes_{1 \leq i \leq r} G^i$ satisfying:

- (1) $|G^i| = N$ for any $1 \leq i \leq r$;
- (2) for any $1 \leq i \leq r$, $G^i = \cup_{1 \leq j \leq \ell_i} H_j^i$, where $H_1^i, H_2^i, \dots, H_{\ell_i}^i$ are symmetric subgraphs of G and $\ell_i \geq 2t$.

Since G_m is F_1 -free, we must have that $|H_1^i| = 1$ for any $1 \leq i \leq r$. That is, G^i consisting of symmetric vertices of G for any $1 \leq i \leq r$.

Let $V' = V(G) - V(\otimes_{1 \leq i \leq r} G^i)$. By symmetry, for each G^i , a vertex $u \in V'$ is adjacent either to all the vertices of G^i , or to no vertex of G^i . Now we divide the vertices in V' by several parts. Let X be the set of vertices u in V' , such that u is adjacent to all the vertices in $\cup_{1 \leq i \leq r} V(G^i)$. For $1 \leq i \leq r$, let X_i be the set of vertices u in V' , such that u is adjacent to all the vertices in $\cup_{1 \leq j \neq i \leq r} V(G^j)$, but to no vertex of $V(G^i)$. Let $Y = V' - (X \cup (\cup_{1 \leq i \leq r} X_i))$. Note that each vertex in Y is adjacent to at most $(r-2)N$ vertices of $\otimes_{1 \leq i \leq r} G^i$. By the definition of q , we must have $|X| \leq q-1$. By a calculation, we have $e(H_m) = e(\mathcal{D}^{-N}(H_m)) + e(T(rN, r)) + (q-1)N + (m-rN)(r-1)N$. Note that

$e(\mathcal{D}^{-N}(H_m)) = e(H_{m-rN})$ by Claim 1. Thus,

$$\begin{aligned}
\phi(G_m) &= e(G_m) - e(H_m) \\
&= e(G_m[V']) + e_G(V(\otimes_{1 \leq i \leq r} G^i), V') + e(T(rN, r)) \\
&\quad - (e(H_{m-rN}) + e(T(rN, r)) + (q-1)N + (m-rN)(r-1)N) \\
&\leq \phi(G_{m-rN}) + e_G(V(\otimes_{1 \leq i \leq r} G^i), V') - ((q-1)N + (m-rN)(r-1)N).
\end{aligned}$$

If $e_G(V(\otimes_{1 \leq i \leq r} G^i), V') - ((q-1)N + (m-rN)(r-1)N) < 0$, then $\phi(G_m) < \phi(G_{m-rN})$, as desired (noting $m-rN \geq \frac{m}{2}$ for large m). Thus, we can assume that

$$e_G(V(\otimes_{1 \leq i \leq r} G^i), V') \geq ((q-1)N + (m-rN)(r-1)N).$$

We will show that G_m has property B . Note that

$$\begin{aligned}
&e_G(V(\otimes_{1 \leq i \leq r} G^i), V') \\
&= e_G(V(\otimes_{1 \leq i \leq r} G^i), X) + e_G(V(\otimes_{1 \leq i \leq r} G^i), \cup_{1 \leq i \leq r} X_i) + e_G(V(\otimes_{1 \leq i \leq r} G^i), Y) \\
&\leq |X|rN + (r-1)N|\cup_{1 \leq i \leq r} X_i| + (r-2)N|Y| \\
&= |X|N + (r-1)N(m-rN) - N|Y| \\
&\leq (q-1)N + (r-1)N(m-rN) - N|Y|.
\end{aligned}$$

Thus, we must have $|X| = q-1$ and $Y = \emptyset$.

For each $1 \leq i \leq r$, let $S_i = V(G^i) \cup X_i$. Then G_m has a partition $V(G_m) = X \cup (\cup_{1 \leq i \leq r} S_i)$ satisfying:

- (3) $|X| = q-1$ and $S_i = V(G^i) \cup X_i$ for any $1 \leq i \leq r$, where $|V(G^i)| = N \geq 2t$;
- (4) for any $1 \leq i \leq r$, $N_{G_m}(v) = V(G) - S_i$ for any $v \in V(G^i)$.

Recall that $F_1 \subseteq tK_2 \otimes T(t(r-1), r-1)$, $F_2 \subseteq (K_{q-1,t} \cup K_{1,t}) \otimes T(t(r-1), r-1)$, and G_m is F_1 -free and F_2 -free. Now we show that, for any $1 \leq i \leq r$, $G_m[X_i]$ is tK_2 -free and $K_{1,t}$ -free. For similarity, we only show this result for $i = 1$. If $G_m[X_1]$ contains a tK_2 , then $G_m[X_1 \cup (\cup_{2 \leq j \leq r} V(G^j))]$ contains a F_1 , a contradiction. If $G_m[X_1]$ contains a $K_{1,t}$, then $G_m[(X \cup S_1) \cup (\cup_{2 \leq j \leq r} V(G^j))]$ contains a F_2 , a contradiction. Hence, $G_m[X_1]$ is tK_2 -free and $K_{1,t}$ -free. It follows that $G_m[X_1]$ will contain no edges after deleting at most t^2 vertices. Let S'_1 be the subset of S_1 after deleting these t^2 vertices. Then $|S_1 - S'_1| \leq t^2$. For any $v \in S'_1 - X_1$, if v is not adjacent to some vertex $w \in V(G_m) - S_1$, let G'' be the graph obtained from G_m by adding all the non-edges between v and $V(G_m) - S_1$. Note that v and the vertices of $V(G^1)$ are symmetric in G'' . By Observation 1, we see that G'' is $\mathcal{F}$ -free. But $e(G'') > e(G_m)$, a contradiction as $G_m \in \text{EX}(m, \mathcal{F})$. Thus $N_{G_m}(v) = V(G_m) - S_1$ for any $v \in S'_1$. For $2 \leq i \leq r$, S'_i is similar to obtain.

Now we show that $||S_i| - |S_j|| \leq 1$ for any $1 \leq i < j \leq r$. Suppose not. Without loss of generality, assume that $|S_1| \geq |S_2| + 2$. let G''' be the graph obtained from G_m

by moving a vertex $v \in S'_1$ to S_2 , i.e., deleting all the edges incident with v in G_m and connecting v to all the vertices in $V(G_m) - S_2$. Note that v and the vertices of S'_2 are symmetric in G''' . Again by Observation 1, we see that G''' is $\mathcal{F}$ -free. But $e(G''') > e(G_m)$, a contradiction as $G_m \in \text{EX}(m, \mathcal{F})$. Thus $||S_i| - |S_j|| \leq 1$ for any $1 \leq i < j \leq r$. It follows that $|S_i| = \lfloor \frac{n-q+1}{r} \rfloor$ or $\lceil \frac{n-q+1}{r} \rceil$ for any $1 \leq i \leq r$. This completes the proof. $\square$

6 Proofs of Theorem 1.5 and Theorem 1.6

In this section, we give the proofs of Theorem 1.5 and Theorem 1.6. We first prove Theorem 1.5.

Proof of Theorem 1.5. (i). Recall that $2 \leq r \leq \ell - 3$ (ℓ is even), and $\mathcal{F} = \{F_1, F_2, F_3\}$, where

$$\begin{aligned} F_1 &= (P_\ell \cup \overline{K_m}) \otimes T(m(r-1), r-1), \\ F_2 &= (K_{1,2} \cup mK_2) \otimes mK_2 \otimes T(m(r-2), r-2), \\ F_3 &= (K_{\frac{\ell}{2}} \cup K_{\ell-1} \cup \overline{K_m}) \otimes T(m(r-1), r-1). \end{aligned}$$

Clearly, $\min_{1 \leq i \leq 3} \chi(F_i) = r+1$. Moreover, for any $1 \leq i \leq 3$, $\chi(F_i - U) \geq r+1$ for any $U \subseteq F_i$ with $|U| = \frac{\ell-2}{2}$. It follows that $K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)$ is $\mathcal{F}$ -free. We will prove that $\text{EX}(n, \mathcal{F}) = \left\{ K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r) \right\}$ for sufficiently large n using the progressive induction.

For $n \geq 1$, let $\mathfrak{U}_n = \text{EX}(n, \mathcal{F})$. For a graph $G' \in \text{EX}(n, \mathcal{F})$, we say G' has property B , if $G' = K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)$. For a graph $G \in \text{EX}(n, \mathcal{F})$, define

$$\phi(G) = e(G) - e(K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)).$$

Clearly, $\phi(G)$ is a non-negative integer-valued function. If G has property B , then $\phi(G) = 0$. By Lemma 3.5, it suffices to show that for sufficiently large n , either G satisfies property B , or there is a $\frac{n}{2} \leq n' < n$ and a $G' \in \text{EX}(n', \mathcal{F})$ such that $\phi(G') > \phi(G)$.

Since $F_1, F_2 \in \mathcal{F}$, we see that $\mathcal{F}$ satisfies the condition in Lemma 1.2 (noting $t = \max_{1 \leq i \leq 3} |F_i|$ there). Since $G \in \text{EX}(n, \mathcal{F})$ with sufficiently large n , there is a constant N (by letting $N_0 = 3m(\ell-1)$ and $N = O(r, t) \cdot N_0$ in Lemma 1.2), such that G has an induced subgraph $\otimes_{1 \leq i \leq r} G^i$ satisfying:

- (1) $|G^i| = N$ for any $1 \leq i \leq r$;
- (2) for any $1 \leq i \leq r$, $G^i = \cup_{1 \leq j \leq \ell_i} H_j^i$, where $H_1^i, H_2^i, \dots, H_{\ell_i}^i$ are symmetric subgraphs of G and $\ell_i \geq 3m$.

Note that $3m(\ell-1)|N|$.

Case 1. There are at least two i , such that $|H_1^i| \geq 2$.

In this case, we can assume that $|H_1^i| \geq 2$ for $i = 1, 2$ without loss of generality. Note that $\ell_i \geq 3m$. Since G is F_2 -free, we must have that $|H_1^1| = |H_1^2| = 2$ and $|H_1^i| \leq 2$ for any $3 \leq i \leq r$. That is, $G^i = \ell_i K_1$ or $G^i = \ell_i K_2$ for any $1 \leq i \leq r$.

Let $V' = V(G) - V(\otimes_{1 \leq i \leq r} G^i)$. By symmetry, for $G^i = \ell_i K_1$, a vertex $u \in V'$ is adjacent either to all the vertices of G^i , or to no vertex of G^i . For $G^i = \ell_i K_2$, a vertex $u \in V'$ is adjacent either to all the vertices of G^i , or to half number of the vertices of G^i , or to no vertex of G^i . Suppose that there is a vertex $v \in V'$ such that v is adjacent to at least $(r-1)N + 1$ vertices of $\otimes_{1 \leq i \leq r} G^i$. Then by symmetry, v must be adjacent to all the vertices of $\otimes_{1 \leq i \neq i_0 \leq r} G^i$ and to at least half number of the vertices of G^{i_0} ($= \ell_{i_0} K_2$) for some $1 \leq i_0 \leq r$. Without loss of generality, assume $i_0 = 1$. Then $G[\{v\} \cup V(G^1 \otimes G^2)]$ contains a $(K_{1,2} \cup mK_2) \otimes mK_2$. And then $G[\{v\} \cup V(\otimes_{1 \leq i \leq r} G^i)]$ contains a F_2 , a contradiction. Thus, for any $v \in V'$, v is adjacent to at most $(r-1)N$ vertices of $\otimes_{1 \leq i \leq r} G^i$. It follows that $e_G(V(\otimes_{1 \leq i \leq r} G^i), V') \leq (n - rN)(r-1)N$.

By a calculation, we have $e(K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)) = e(K_{\frac{\ell-2}{2}} \otimes T(n - rN - \frac{\ell-2}{2}, r)) + e(T(rN, r)) + \frac{\ell-2}{2}N + (n - rN)(r-1)N$. Let $G' \in \text{EX}(n - rN, \mathcal{F})$. Then $e(G[V']) \leq e(G')$. Note that $\phi(G') = e(G') - e(K_{\frac{\ell-2}{2}} \otimes T(n - rN - \frac{\ell-2}{2}, r))$. Thus,

$$\begin{aligned}
\phi(G) &= e(G) - e(K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)) \\
&\leq e(G[V']) + e_G(V(\otimes_{1 \leq i \leq r} G^i), V') + e(T(rN, r)) + \frac{N}{2}r \\
&\quad - e(K_{\frac{\ell-2}{2}} \otimes T(n - rN - \frac{\ell-2}{2}, r)) - e(T(rN, r)) - \frac{\ell-2}{2}N - (n - rN)(r-1)N \\
&\leq \phi(G') + (n - rN)(r-1)N + \frac{N}{2}r - \frac{\ell-2}{2}N - (n - rN)(r-1)N \\
&= \phi(G') - \frac{N}{2}(\ell - 2 - r) \text{ (noting } r \leq \ell - 3) \\
&< \phi(G'),
\end{aligned}$$

as desired.

Case 2. There is at most one i , such that $|H_1^i| \geq 2$.

In this case, we can assume that $|H_1^i| = 1$ for $i \geq 2$ without loss of generality. That is $G_i = N \cdot K_1$ for $i \geq 2$. Recall that $3m(\ell-1) \mid N$. Also, let $V' = V(G) - V(\otimes_{1 \leq i \leq r} G^i)$. By symmetry, we can divide the vertices in V' by $r+1$ parts as follows. Let X_1 be the set of vertices u in V' , such that u is adjacent to all the vertices of $\otimes_{2 \leq i \leq r} G^i$. For $2 \leq i \leq r$, let X_i be the set of vertices u in V' , such that u is adjacent to all the vertices of $\otimes_{1 \leq j \neq i \leq r} G^j$ but to no vertex of $V(G^i)$. Let $Y = V' - \cup_{1 \leq i \leq r} X_i$, i.e., the set of vertices which have non-neighbors in at least two among G^i with $1 \leq i \leq r$. Note that each vertex in $\cup_{2 \leq i \leq r} X_i$ is adjacent to $(r-1)N$ vertices of $\otimes_{1 \leq i \leq r} G^i$. Each vertex in Y is adjacent to at most

$(r-1)N-1$ vertices of $\otimes_{1 \leq i \leq r} G^i$ (using symmetry). The way connecting the vertices in X_1 to $V(G^1)$ is not known now.

Recall that $e(K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)) = e(K_{\frac{\ell-2}{2}} \otimes T(n - rN - \frac{\ell-2}{2}, r)) + e(T(rN, r)) + \frac{\ell-2}{2}N + (n - rN)(r-1)N$. Let $G' \in \text{EX}(n - rN, \mathcal{F})$. Then $e(G[V']) \leq e(G')$. Note that $\phi(G') = e(G') - e(K_{\frac{\ell-2}{2}} \otimes T(n - rN - \frac{\ell-2}{2}, r))$. Thus,

$$\begin{aligned} \phi(G) &= e(G) - e(K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)) \\ &= e(G[V']) + e_G(V(\otimes_{1 \leq i \leq r} G^i), V') + e(T(rN, r)) + e(G^1) \\ &\quad - e(K_{\frac{\ell-2}{2}} \otimes T(n - rN - \frac{\ell-2}{2}, r)) - e(T(rN, r)) - \frac{\ell-2}{2}N - (n - rN)(r-1)N \\ &\leq \phi(G') + e_G(V(\otimes_{1 \leq i \leq r} G^i), V') + e(G^1) - \frac{\ell-2}{2}N - (n - rN)(r-1)N. \end{aligned}$$

If $e_G(V(\otimes_{1 \leq i \leq r} G^i), V') + e(G^1) - \frac{\ell-2}{2}N - (n - rN)(r-1)N < 0$, then $\phi(G) < \phi(G')$, as desired. Thus, we can assume that

$$e_G(V(\otimes_{1 \leq i \leq r} G^i), V') + e(G^1) \geq \frac{\ell-2}{2}N + (n - rN)(r-1)N.$$

We will show that G has property B (i.e., $G = K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)$).

Note that

$$\begin{aligned} e_G(V(\otimes_{1 \leq i \leq r} G^i), V') &= e_G(V(\otimes_{1 \leq i \leq r} G^i), X_1) + e_G(V(\otimes_{1 \leq i \leq r} G^i), \cup_{2 \leq i \leq r} X_i) + e_G(V(\otimes_{1 \leq i \leq r} G^i), Y) \\ &\leq e_G(V(\otimes_{1 \leq i \leq r} G^i), X_1) + (r-1)N|\cup_{2 \leq i \leq r} X_i| + ((r-1)N-1)|Y| \\ &= e_G(V(\otimes_{1 \leq i \leq r} G^i), X_1) + (r-1)N \cdot (n - rN - |X_1|) - |Y| \\ &= e_G(V(G^1), X_1) + (r-1)N|X_1| + (r-1)N \cdot (n - rN - |X_1|) - |Y| \\ &= e_G(V(G^1), X_1) + (r-1)N \cdot (n - rN) - |Y|. \end{aligned}$$

Recall $e_G(V(\otimes_{1 \leq i \leq r} G^i), V') + e(G^1) \geq \frac{\ell-2}{2}N + (n - rN)(r-1)N$. It follows that

$$e(G^1) + e_G(V(G^1), X_1) \geq \frac{\ell-2}{2}N + |Y| \geq \frac{\ell-2}{2}N.$$

Note that each vertex in $V(G^1) \cup X_1$ is adjacent to all the vertices in $V(\otimes_{2 \leq i \leq r} G^i)$. Thus, $G[V(G^1) \cup X_1]$ is P_ℓ -free, since G is F_1 -free. Note that $H_1^1, H_1^1, \dots, H_{\ell_1}^1$ are also symmetric in $G[V(G^1) \cup X_1]$ and $|G^1| = N$. By Lemma 3.7, we have $e(G^1) + e_G(V(G^1), X_1) = \frac{\ell-2}{2}N$ (implying $|Y| = 0$). Moreover, either $Q_i^1 = K_{\ell-1}$ for any $1 \leq i \leq \ell_1$ and $e_G(V(G^1), X_1) = 0$, or $Q_i^1 = K_1$ for any $1 \leq i \leq \ell_1$ and there are $\frac{\ell-2}{2}$ vertices in X_1 which are all adjacent to the (unique) vertex in Q_i^1 for all $1 \leq i \leq \ell_1$. If $Q_i^1 = K_{\ell-1}$ for any $1 \leq i \leq \ell_1$, then F_3 is contained in G , a contradiction. Hence, we must have the latter case. That is, $G^1 = N \cdot K_1$,

and there are $\frac{\ell-2}{2}$ vertices in X_1 , say $y_1, y_2, \dots, y_{\frac{\ell-2}{2}}$, such that they are all adjacent to all the vertices of G^1 . Recall $Y = \emptyset$. Let $X = \{y_1, y_2, \dots, y_{\frac{\ell-2}{2}}\}$. For convenience, we still use X_1 to denote $X_1 - X$. For each $1 \leq i \leq r$, let $S_i = V(G^i) \cup X_i$. Then G has a partition $V(G) = X \cup (\cup_{1 \leq i \leq r} S_i)$ satisfying:

- (3) $X = \{y_1, y_2, \dots, y_{\frac{\ell-2}{2}}\}$ and $S_i = V(G^i) \cup X_i$ for any $1 \leq i \leq r$, where $|V(G^i)| = N \geq 3m$;
- (4) for any $1 \leq i \leq r$, $N_G(v) = V(G) - S_i$ for any $v \in V(G^i)$.

For any $1 \leq i \leq r$, let X'_i be the set of vertices in S_i which have neighbors in X . Now we show that the vertices in X'_i are isolated in $G[S_i]$. Suppose not. Without loss of generality, assume that $v \in X'_1$ such that v is adjacent to $y_{\frac{\ell-2}{2}}$ and adjacent to another vertex w in S_1 . Let $u_1, u_2, \dots, u_{\frac{\ell-2}{2}}$ be other $\frac{\ell-2}{2}$ vertices (not v, w) in $V(G^1)$. Then $u_1 y_1 u_2 y_2 \cdots u_{\frac{\ell-2}{2}} y_{\frac{\ell-2}{2}} v w$ is a P_ℓ . Since the vertices of this path are all adjacent to the vertices in $V(\otimes_{2 \leq i \leq r} G^i)$, we see F_1 is contained in G , a contradiction. Hence, the vertices in X'_i are isolated in $G[S_i]$ for any $1 \leq i \leq r$.

Note that $G[X_i - X'_i]$ is P_ℓ -free for any $1 \leq i \leq r$. By Lemma 3.3, we have $e(G[X_i - X'_i]) \leq \frac{\ell-2}{2}|X_i - X'_i|$ with equality if and only if $G[X_i - X'_i]$ is the disjoint union of several copies of $K_{\ell-1}$. Thus,

$$\begin{aligned} e(G) &\leq e(K_{\frac{\ell-2}{2}} \otimes K_{|S_1|, |S_2|, \dots, |S_r|}) + (\sum_{1 \leq i \leq r} e(G[X_i - X'_i])) - (\sum_{1 \leq i \leq r} \frac{\ell-2}{2}|X_i - X'_i|) \\ &\leq e(K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)) + (\sum_{1 \leq i \leq r} e(G[X_i - X'_i])) - (\sum_{1 \leq i \leq r} \frac{\ell-2}{2}|X_i - X'_i|) \\ &\leq e(K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)). \end{aligned}$$

However, $e(G) \geq e(K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r))$ as $G \in \text{EX}(n, \mathcal{F})$. This forces that $G[X] = K_{\frac{\ell-2}{2}}$, and $||S_i| - \frac{n - \frac{\ell-2}{2}}{r}| < 1$ and $e(G[X_i - X'_i]) = \frac{\ell-2}{2}|X_i - X'_i|$ for any $1 \leq i \leq r$. Then $G[X_i - X'_i]$ is the disjoint union of $K_{\ell-1}$. If $X_i - X'_i$ is not empty for some $1 \leq i \leq r$, say $i = 1$ without loss of generality, then $G[X_1 - X'_1]$ contains a $K_{\ell-1}$. Now we choose a vertex u_1 in $V(G^1)$. Then $G[\{u_1, y_1, y_2, \dots, y_{\frac{\ell-2}{2}}\}] = K_{\frac{\ell}{2}}$. Clearly, $G[X \cup S_1]$ contains a $K_{\frac{\ell}{2}} \cup K_{\ell-1} \cup \overline{K_m}$. Since all vertices of $\otimes_{2 \leq i \leq r} G^i$ are adjacent to vertices in $X \cup S_1$, G contains a F_3 , a contradiction. Hence, $X_i - X'_i = \emptyset$ for any $1 \leq i \leq r$. This means that all the vertices in S_i are isolated in $G[S_i]$ for any $1 \leq i \leq r$. Then, G is a spanning subgraph of $K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)$. Hence, $G = K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)$, as desired.

(ii). Ignore the symbols used in (i). Recall that $2 \leq r = \ell - 2$ is even, and $\mathcal{F} = \{F_1, F_2, F_3\}$, where

$$\begin{aligned} F_1 &= (P_{r+2} \cup \overline{K_m}) \otimes T(m(r-1), r-1), \\ F_2 &= (K_{1,2} \cup mK_2) \otimes mK_2 \otimes T(m(r-2), r-2), \end{aligned}$$

$$F_3 = (K_{\frac{r+2}{2}} \cup K_{r+1} \cup \overline{K_m}) \otimes T(m(r-1), r-1).$$

Clearly, $\min_{1 \leq i \leq 3} \chi(F_i) = r+1$. Moreover, for any $1 \leq i \leq 3$, $\chi(F_i - U) \geq r+1$ for any $U \subseteq F_i$ with $|U| = \frac{r}{2}$. It follows that $K_{\frac{r}{2}} \otimes T(n - \frac{r}{2}, r)$ is $\mathcal{F}$ -free. For $n \geq 2r$, recall that $G_{n,r}$ is the graph obtained from $T(n, r)$ by adding a (near) perfect matching in each of its r parts. By a result in [19], we see that $G_{n,r}$ is $K_1 \otimes (\otimes_{1 \leq i \leq r} 2K_1)$ -free. Clearly, $K_1 \otimes (\otimes_{1 \leq i \leq r} 2K_1)$ is contained in F_i for any $1 \leq i \leq 3$. Thus, $G_{n,r}$ is $\mathcal{F}$ -free. We will prove that $\text{EX}(n, \mathcal{F}) \subseteq \{K_{\frac{r}{2}} \otimes T(n - \frac{r}{2}, r)\} \cup \mathcal{G}_{n,r}$ for sufficiently large n using the progressive induction.

For $n \geq 1$, let $\mathfrak{U}_n = \text{EX}(n, \mathcal{F})$. For a graph $G' \in \text{EX}(n, \mathcal{F})$, we say G' has property B , if $G' \in \{K_{\frac{r}{2}} \otimes T(n - \frac{r}{2}, r)\} \cup \mathcal{G}_{n,r}$. For a graph $G \in \text{EX}(n, \mathcal{F})$, define

$$\phi(G) = \min \left\{ e(G) - e(K_{\frac{r}{2}} \otimes T(n - \frac{r}{2}, r)), e(G) - e(G_{n,r}) \right\}.$$

Clearly, $\phi(G)$ is a non-negative integer-valued function. If G has property B , then $\phi(G) = 0$. By Lemma 3.5, it suffices to show that for sufficiently large n , either G satisfies property B , or there is a $\frac{n}{2} \leq n' < n$ and a $G' \in \text{EX}(n', \mathcal{F})$ such that $\phi(G') > \phi(G)$.

For even integer N , it is clear that (for large n) $e(G_{n,r}) = e(G_{n-2N,r}) + e(T(rN, r)) + \frac{rN}{2} + (n-rN)(r-1)N$, and $e(K_{\frac{r}{2}} \otimes T(n - \frac{r}{2}, r)) = e(K_{\frac{r}{2}} \otimes T(n - rN - \frac{r}{2}, r)) + e(T(rN, r)) + \frac{r}{2}N + (n-rN)(r-1)N$. For $G \in \text{EX}(n, \mathcal{F})$ and $G' \in \text{EX}(n-rN, \mathcal{F})$, it follows that

$$\begin{aligned} \phi(G) &= \min \left\{ e(G) - e(K_{\frac{r}{2}} \otimes T(n - \frac{r}{2}, r)), e(G) - e(G_{n,r}) \right\} \\ &= e(G) - e(G') + \min \left\{ e(G') - e(K_{\frac{r}{2}} \otimes T(n - \frac{r}{2}, r)), e(G') - e(G_{n,r}) \right\} \\ &= e(G) - e(G') - (e(T(rN, r)) + \frac{r}{2}N + (n-rN)(r-1)N) \\ &\quad + \min \left\{ e(G') - e(K_{\frac{r}{2}} \otimes T(n - rN - \frac{r}{2}, r)), e(G') - e(G_{n-rN,r}) \right\} \\ &= \phi(G') + e(G) - e(G') - (e(T(rN, r)) + \frac{r}{2}N + (n-rN)(r-1)N). \end{aligned}$$

If $e(G) - e(G') - (e(T(rN, r)) + \frac{r}{2}N + (n-rN)(r-1)N) < 0$, then $\phi(G) < \phi(G')$, as desired. Thus, we can assume

$$e(G) \geq e(G') + e(T(rN, r)) + \frac{r}{2}N + (n-rN)(r-1)N$$

in the following. We will show that G has property B (i.e., $G \in \{K_{\frac{r}{2}} \otimes T(n - \frac{r}{2}, r)\} \cup \mathcal{G}_{n,r}$). Recall that $G' \in \text{EX}(n-rN, \mathcal{F})$.

Since $F_1, F_2 \in \mathcal{F}$, we see that $\mathcal{F}$ satisfies the condition in Lemma 1.2. Since $G \in \text{EX}(n, \mathcal{F})$ with sufficiently large n , there is a constant N (by letting $N_0 = 4m$ and $N = O(r, t) \cdot N_0$ in Lemma 1.2), such that G has an induced subgraph $\otimes_{1 \leq i \leq r} G^i$ satisfying:

(1) $|G^i| = N$ for any $1 \leq i \leq r$;

(2) for any $1 \leq i \leq r$, $G^i = \cup_{1 \leq j \leq \ell_i} H_j^i$, where $H_1^i, H_2^i, \dots, H_{\ell_i}^i$ are symmetric subgraphs of G and $\ell_i \geq 4m$.

Note that $4m|N$.

Case 1. There are at least two i , such that $|H_1^i| \geq 2$.

In this case, we can assume that $|H_1^i| \geq 2$ for $i = 1, 2$ without loss of generality. Note that $\ell_i \geq 4m$. Since G is F_2 -free, we must have that $|H_1^1| = |H_1^2| = 2$ and $|H_1^i| \leq 2$ for any $3 \leq i \leq r$. Let $V' = V(G) - V(\otimes_{1 \leq i \leq r} G^i)$. Suppose that there is a vertex $v \in V'$ such that v is adjacent to at least $(r-1)N + 1$ vertices of $\otimes_{1 \leq i \leq r} G^i$. Then by symmetry, v must be adjacent to all the vertices of $\otimes_{1 \leq i \neq i_0 \leq r} G^i$ and to at least half number of the vertices of G^{i_0} for some $1 \leq i_0 \leq 2$. Clearly, $G[\{v\} \cup V(\otimes_{1 \leq i \leq r} G^i)]$ contains a F_2 , a contradiction. Thus, for any $v \in V'$, v is adjacent to at most $(r-1)N$ vertices of $\otimes_{1 \leq i \leq r} G^i$. Let Y be the set of vertices $v \in V'$, such that v is adjacent to at most $(r-1)N - 1$ vertices of $\otimes_{1 \leq i \leq r} G^i$. It follows that $e_G(V(\otimes_{1 \leq i \leq r} G^i), V') \leq (n - rN)(r-1)N - |Y|$.

Since $G[V']$ is $\mathcal{F}$ -free, we have $e(G[V']) \leq e(G')$ as $G' \in \text{EX}(n - rN, \mathcal{F})$. Recall that $e(G) \geq e(G') + e(T(rN, r)) + \frac{r}{2}N + (n - rN)(r-1)N$. Thus, $e(G) \geq e(G[V']) + e(T(rN, r)) + \frac{r}{2}N + (n - rN)(r-1)N$. Note that $e(\otimes_{1 \leq i \leq r} G^i) \leq \frac{rN}{2} + e(T(rN, r))$. However,

$$\begin{aligned} e(G) &= e(G[V']) + e_G(V(\otimes_{1 \leq i \leq r} G^i), V') + e(\otimes_{1 \leq i \leq r} G^i) \\ &\leq e(G[V']) + (n - rN)(r-1)N - |Y| + \frac{rN}{2} + e(T(rN, r)). \end{aligned}$$

We must have $|Y| = 0$, $e(\otimes_{1 \leq i \leq r} G^i) = \frac{rN}{2} + e(T(rN, r))$, and each vertex in V' is adjacent to exactly $(r-1)N$ vertices of $\otimes_{1 \leq i \leq r} G^i$. It follows that $G^i = \frac{N}{2}K_2$ for any $1 \leq i \leq r$. If some vertex $v \in V'$ has neighbors in $V(G^i)$ for each $1 \leq i \leq r$, then by symmetry, G contains F_1 for $r = 2$, and G contains F_2 for $r \geq 4$ (r is even), a contradiction. Hence, each vertex $v \in V'$ is adjacent to no vertex of $V(G^i)$ for some $1 \leq i \leq r$, and thus adjacent to all vertices of $\otimes_{1 \leq j \neq i \leq r} G^j$. Then by the above discussion, $V' = \cup_{1 \leq i \leq r} X_i$, where X_i is adjacent to exactly the vertices of $\otimes_{1 \leq j \neq i \leq r} G^j$. Since G is F_2 -free, $G[X_i]$ has no vertex of degree at least 2 for any $1 \leq i \leq r$. We must have $G \in \mathcal{G}_{n,r}$ as G has maximum edges, as desired.

Case 2. There is at most one i , such that $|H_1^i| \geq 2$.

In this case, we can assume that $|H_1^i| = 1$ for $i \geq 2$ without loss of generality. That is $G_i = N \cdot K_1$ for $i \geq 2$. Recall that $4m|N$. Also, let $V' = V(G) - V(\otimes_{1 \leq i \leq r} G^i)$. By symmetry, we can divide the vertices in V' by $r+1$ parts as follows. Let X_1 be the set of vertices u in V' , such that u is adjacent to all the vertices of $\otimes_{2 \leq i \leq r} G^i$. For $2 \leq i \leq r$, let X_i be the set of vertices u in V' , such that u is adjacent to all the vertices of $\otimes_{1 \leq j \neq i \leq r} G^j$ but to no vertex of $V(G^i)$. Let $Y = V' - \cup_{1 \leq i \leq r} X_i$, i.e., the set of vertices which have non-neighbors in at least two among G^i with $1 \leq i \leq r$. Note that each vertex in $\cup_{2 \leq i \leq r} X_i$

is adjacent to $(r-1)N$ vertices of $\otimes_{1 \leq i \leq r} G^i$. Each vertex in Y is adjacent to at most $(r-1)N-1$ vertices of $\otimes_{1 \leq i \leq r} G^i$ (using symmetry). The way connecting the vertices in X_1 to $V(G^1)$ is not known now. Note that

$$\begin{aligned}
e(G) &= e(G[V']) + e(T(rN, r)) + e(G^1) + e_G(V(\otimes_{1 \leq i \leq r} G^i), V') \\
&= e(G[V']) + e(T(rN, r)) + e(G^1) \\
&\quad + e_G(V(\otimes_{1 \leq i \leq r} G^i), X_1) + e_G(V(\otimes_{1 \leq i \leq r} G^i), \cup_{2 \leq i \leq r} X_i) + e_G(V(\otimes_{1 \leq i \leq r} G^i), Y) \\
&\leq e(G[V']) + e(T(rN, r)) + e(G^1) \\
&\quad + e_G(V(\otimes_{1 \leq i \leq r} G^i), X_1) + (r-1)N|\cup_{2 \leq i \leq r} X_i| + ((r-1)N-1)|Y| \\
&= e(G[V']) + e(T(rN, r)) + e(G^1) \\
&\quad + e_G(V(\otimes_{1 \leq i \leq r} G^i), X_1) + (r-1)N \cdot (n-rN-|X_1|) - |Y| \\
&= e(G[V']) + e(T(rN, r)) + e(G^1) \\
&\quad + e_G(V(G^1), X_1) + (r-1)N|X_1| + (r-1)N \cdot (n-rN-|X_1|) - |Y| \\
&= e(G[V']) + e(G^1) + e_G(V(G^1), X_1) + e(T(rN, r)) + (r-1)N \cdot (n-rN) - |Y|.
\end{aligned}$$

We will show that $G = K_{\frac{r}{2}} \otimes T(n - \frac{r}{2}, r)$. Recall that

$$e(G) \geq e(G') + e(T(rN, r)) + \frac{r}{2}N + (n-rN)(r-1)N,$$

where $G' \in \text{EX}(n-rN, \mathcal{F})$. Since the rest of the proof is very similar to the Case 2 in (i), we omit it. This completes the proof. $\square$

Proof of Theorem 1.6. Let

$$F^1 = P_6 \otimes 3K_1 \otimes 3K_1,$$

$$F^2 = (K_{1,2} \cup K_2) \otimes (K_2 \cup 2K_1) \otimes 3K_1,$$

and

$$F^3 = 2K_3 \otimes 3K_1 \otimes 3K_1.$$

Then, (see [21]) $I^{12} \subseteq F^i$ for any $1 \leq i \leq 3$. Thus, $mI^{12} \subseteq mF^i$ for any $1 \leq i \leq 3$. Recall that $\chi(I^{12}) = 4$ and $\chi(I^{12} - U) = 4$ for any $U \subseteq V(I^{12})$ with $|U| = 2$. Thus, $\chi(mI^{12} - U) = 4$ for any $U \subseteq V(mI^{12})$ with $|U| = 3m-1$. It follows that $K_{3m-1} \otimes T(n-3m+1, 3)$ is mI^{12} -free. We will prove that $\text{EX}(n, mI^{12}) = \{K_{3m-1} \otimes T(n-3m+1, 3)\}$ for sufficiently large n using the progressive induction.

For $n \geq 1$, let $\mathfrak{U}_n = \text{EX}(n, mI^{12})$. For a graph $G' \in \text{EX}(n, mI^{12})$, we say G' has property B , if $G' = K_{3m-1} \otimes T(n-3m+1, 3)$. For a graph $G \in \text{EX}(n, mI^{12})$, define

$$\phi(G) = e(G) - e(K_{3m-1} \otimes T(n-3m+1, 3)).$$

Clearly, $\phi(G)$ is a non-negative integer-valued function. If G has property B , then $\phi(G) = 0$. By Lemma 3.5, it suffices to show that for sufficiently large n , either G satisfies property B , or there is a $\frac{n}{2} \leq n' < n$ and a $G' \in \text{EX}(n', mI^{12})$ such that $\phi(G') > \phi(G)$.

Let $G \in \text{EX}(n, mI^{12})$ with sufficiently large n . Recall that $mI^{12} \subseteq mF^1, mF^2$. By Lemma 1.2 (noting $t = 12m$ there), there is a constant N (by letting $N_0 = 12(6m - 1)$ and $N = O(r, t) \cdot N_0$ in Lemma 1.2), such that G has an induced subgraph $\otimes_{1 \leq i \leq 3} G^i$ satisfying:

- (1) $|G^i| = N$ for any $1 \leq i \leq 3$;
- (2) for any $1 \leq i \leq 3$, $G^i = \cup_{1 \leq j \leq \ell_i} H_j^i$, where $H_1^i, H_2^i, \dots, H_{\ell_i}^i$ are symmetric subgraphs of G and $\ell_i \geq 12(6m - 1)$.

Note that $12(6m - 1) |N$.

Case 1. There are at least two i , such that $|H_1^i| \geq 2$.

In this case, we can assume that $|H_1^i| \geq 2$ for $i = 1, 2$ without loss of generality. Note that $\ell_i \geq 12m$. Since G is mF^2 -free, we must have that $|H_1^1| = |H_1^2| = 2$ and $|H_1^3| \leq 2$. That is, $G^i = \ell_i K_1$ or $G^i = \ell_i K_2$ for any $1 \leq i \leq 3$.

Let $V' = V(G) - V(\otimes_{1 \leq i \leq 3} G^i)$. Let X be the set of vertices $u \in V'$ such that u has at least $2N + 1$ neighbors in $\otimes_{1 \leq i \leq 3} G^i$. By symmetry, any vertex $u \in X$ has at most $\frac{N}{2}$ non-neighbors in $\otimes_{1 \leq i \leq 3} G^i$. It is easy to see that F^2 is a subgraph of G induced by u and $\cup_{1 \leq i \leq 3, 1 \leq j \leq 6} V(H_j^i)$. If $|X| \geq m$, assume that $u_\ell \in X$ with $1 \leq \ell \leq m$. Then mF^2 is a subgraph of G induced by $\{u_1, u_2, \dots, u_m\}$ and $\cup_{1 \leq i \leq 3, 1 \leq j \leq 6m} V(H_j^i)$. This contradicts that G is mF^2 -free. Hence, $|X| \leq m - 1$. Since $v \in V' - X$ is adjacent to at most $2N$ vertices of $\otimes_{1 \leq i \leq 3} G^i$, we have

$$e_G(V(\otimes_{1 \leq i \leq 3} G^i), V') \leq (n - 3N)2N + |X|N \leq (n - 3N)2N + (m - 1)N.$$

By a calculation, we have $e(K_{3m-1} \otimes T(n - 3m + 1, 3)) = e(K_{3m-1} \otimes T(n - 3N - 3m + 1, 3)) + e(T(3N, 3)) + (3m - 1)N + (n - 3N)2N$. Let $G' \in \text{EX}(n - rN, \mathcal{F})$. Then $e(G[V']) \leq e(G')$. Note that $\phi(G') = e(G') - e(K_{3m-1} \otimes T(n - 3N - 3m + 1, 3))$. Thus,

$$\begin{aligned} \phi(G) &= e(G) - e(K_{3m-1} \otimes T(n - 3m + 1, 3)) \\ &\leq e(G[V']) + e_G(V(\otimes_{1 \leq i \leq 3} G^i), V') + e(T(3N, 3)) + \frac{3N}{2} \\ &\quad - e(K_{3m-1} \otimes T(n - 3N - 3m + 1, 3)) - e(T(3N, 3)) - (3m - 1)N - (n - 3N)2N \\ &\leq \phi(G') + (n - 3N)2N + (m - 1)N + \frac{3N}{2} - (3m - 1)N - (n - 3N)2N \\ &= \phi(G') - \frac{N}{2}(4m - 3) \\ &< \phi(G'), \end{aligned}$$

as desired.

Case 2. There is at most one i , such that $|H_1^i| \geq 2$.

In this case, we can assume that $|H_1^2| = |H_1^3| = 1$ without loss of generality. That is $G_i = N \cdot K_1$ for $i = 2, 3$. Recall that $12(6m-1)|N$. Let $V' = V(G) - V(\otimes_{1 \leq i \leq 3} G^i)$. By symmetry, we can divide the vertices in V' by 4 parts as follows. Let X_1 be the set of vertices u in V' , such that u is adjacent to all the vertices of $\otimes_{2 \leq i \leq 3} G^i$. For $2 \leq i \leq 3$, let X_i be the set of vertices u in V' , such that u is adjacent to all the vertices of $\otimes_{1 \leq j \neq i \leq 3} G^j$ but to no vertex of $V(G^i)$. Let $Y = V' - \cup_{1 \leq i \leq 3} X_i$. Note that each vertex in $\cup_{2 \leq i \leq 3} X_i$ is adjacent to $2N$ vertices of $\otimes_{1 \leq i \leq 3} G^i$. Each vertex in Y is adjacent to at most $2N - 1$ vertices of $\otimes_{1 \leq i \leq 3} G^i$ (using symmetry).

Recall that $e(K_{3m-1} \otimes T(n - 3m + 1, 3)) = e(K_{3m-1} \otimes T(n - 3N - 3m + 1, 3)) + e(T(3N, 3)) + (3m-1)N + (n-3N)2N$. Let $G' \in \text{EX}(n-3N, mI^{12})$. Then $e(G[V']) \leq e(G')$. Note that $\phi(G') = e(G') - e(K_{3m-1} \otimes T(n - 3N - 3m + 1, 3))$. Thus,

$$\begin{aligned} \phi(G) &= e(G) - e(K_{3m-1} \otimes T(n - 3m + 1, 2)) \\ &= e(G[V']) + e_G(V(\otimes_{1 \leq i \leq 3} G^i), V') + e(T(3N, 3)) + e(G^1) \\ &\quad - e(K_{3m-1} \otimes T(n - 3N - 3m + 1, 3)) - e(T(3N, 3)) - (3m-1)N - (n-3N)2N \\ &\leq \phi(G') + e_G(V(\otimes_{1 \leq i \leq 3} G^i), V') + e(G^1) - (3m-1)N - (n-3N)2N. \end{aligned}$$

If $e_G(V(\otimes_{1 \leq i \leq 3} G^i), V') + e(G^1) - (3m-1)N - (n-3N)2N < 0$, then $\phi(G) < \phi(G')$, as desired. Thus, we can assume that

$$e_G(V(\otimes_{1 \leq i \leq 3} G^i), V') + e(G^1) \geq (3m-1)N + (n-3N)2N.$$

We will show that G has property B (i.e., $G = K_{3m-1} \otimes T(n - 3m + 1, 3)$).

Note that

$$\begin{aligned} &e_G(V(\otimes_{1 \leq i \leq 3} G^i), V') \\ &= e_G(V(\otimes_{1 \leq i \leq 3} G^i), X_1) + e_G(V(\otimes_{1 \leq i \leq 3} G^i), \cup_{2 \leq i \leq 3} X_i) + e_G(V(\otimes_{1 \leq i \leq 3} G^i), Y) \\ &\leq e_G(V(\otimes_{1 \leq i \leq 3} G^i), X_1) + 2N|\cup_{2 \leq i \leq 3} X_i| + (2N-1)|Y| \\ &= e_G(V(\otimes_{1 \leq i \leq 3} G^i), X_1) + 2N \cdot (n-3N - |X_1|) - |Y| \\ &= e_G(V(G^1), X_1) + 2N|X_1| + 2N \cdot (n-3N - |X_1|) - |Y| \\ &= e_G(V(G^1), X_1) + 2N \cdot (n-3N) - |Y|. \end{aligned}$$

Recall

$$e_G(V(\otimes_{1 \leq i \leq 3} G^i), V') + e(G^1) \geq (3m-1)N + (n-3N)2N.$$

It follows that

$$e(G^1) + e_G(V(G^1), X_1) \geq (3m-1)N + |Y| \geq (3m-1)N.$$

Note that each vertex in $V(G^1) \cup X_1$ is adjacent to all the vertices in $V(\otimes_{2 \leq i \leq 3} G^i)$. Thus, $G[V(G^1) \cup X_1]$ is P_{6m} -free, since G is mF^1 -free. Note that $H_1^1, H_1^1, \dots, H_{\ell_1}^1$ are also symmetric in $G[V(G^1) \cup X_1]$ and $|G^1| = N$. By Lemma 3.7, we have $e(G^1) + e_G(V(G^1), X_1) = (3m - 1)N$ (implying $|Y| = 0$). Moreover, either $Q_i^1 = K_{6m-1}$ for any $1 \leq i \leq \ell_1$ and $e_G(V(G^1), X_1) = 0$, or $Q_i^1 = K_1$ for any $1 \leq i \leq \ell_1$ and there are $3m - 1$ vertices in X_1 which are all adjacent to the (unique) vertex in Q_i^1 for all $1 \leq i \leq \ell_1$. If $Q_i^1 = K_{6m-1}$ for any $1 \leq i \leq \ell_1$, then mF^3 is contained in $\otimes_{1 \leq i \leq 3} G^i$, a contradiction. Hence, we must have the latter case. That is, $G^1 = N \cdot K_1$, and there are $3m - 1$ vertices in X_1 , say $y_1, y_2, \dots, y_{3m-1}$, such that they are all adjacent to all the vertices of G^1 . Recall $Y = \emptyset$. Let $X = \{y_1, y_2, \dots, y_{3m-1}\}$. For convenience, we still use X_1 to denote $X_1 - X$. For each $1 \leq i \leq 3$, let $S_i = V(G^i) \cup X_i$. Then G has a partition $V(G) = X \cup (\cup_{1 \leq i \leq 3} S_i)$ satisfying:

- (3) $X = \{y_1, y_2, \dots, y_{3m-1}\}$ and $S_i = V(G^i) \cup X_i$ for any $1 \leq i \leq 3$, where $|V(G^i)| = N \geq 12m$;
- (4) for any $1 \leq i \leq 3$, $N_G(v) = V(G) - S_i$ for any $v \in V(G^i)$.

For any $1 \leq i \leq 3$, let X'_i be the set of vertices in S_i which have neighbors in X . Now we show that the vertices in X'_i are isolated in $G[S_i]$. Suppose not. Without loss of generality, assume that $v \in X'_1$ such that v is adjacent to y_{3m-1} and adjacent to another vertex w in S_1 . Let $u_1, u_2, \dots, u_{3m-1}$ be other $3m - 1$ vertices in $V(G^1)$. Then $u_1 y_1 u_2 y_2 \cdots u_{3m-1} y_{3m-1} v w$ is a P_{6m} . Since the vertices of this path are all adjacent to the vertices in $V(\otimes_{2 \leq i \leq 3} G^i)$, we see mF^1 is contained in G , a contradiction. Hence, the vertices in X'_i are isolated in $G[S_i]$ for any $1 \leq i \leq 3$.

Note that $G[X_i - X'_i]$ is P_{6m} -free for any $1 \leq i \leq 3$. By Lemma 3.3, we have $e(G[X_i - X'_i]) \leq (3m - 1)|X_i - X'_i|$ with equality if and only if $G[X_i - X'_i]$ is the disjoint union of K_{6m-1} . Thus,

$$\begin{aligned}
e(G) &\leq e(K_{3m-1} \otimes K_{|S_1|, |S_2|, \dots, |S_3|}) + \left(\sum_{1 \leq i \leq 3} e(G[X_i - X'_i]) \right) - \left(\sum_{1 \leq i \leq 3} (3m - 1)|X_i - X'_i| \right) \\
&\leq e(K_{3m-1} \otimes T(n - 3m + 1, 3)) + \left(\sum_{1 \leq i \leq 3} e(G[X_i - X'_i]) \right) - \left(\sum_{1 \leq i \leq 3} (3m - 1)|X_i - X'_i| \right) \\
&\leq e(K_{3m-1} \otimes T(n - 3m + 1, 3)).
\end{aligned}$$

However, $e(G) \geq e(K_{3m-1} \otimes T(n - 3m + 1, 3))$. This forces that $G[X] = K_{3m-1}$, and $||S_i| - \frac{n-3m+1}{3}| < 1$ and $e(G[X_i - X'_i]) = (3m - 1)|X_i - X'_i|$ for any $1 \leq i \leq 3$. Then $G[X_i - X'_i]$ is the disjoint union of K_{6m-1} . If $X_i - X'_i$ is not empty for some $1 \leq i \leq 3$, say $i = 1$ without loss of generality, then $G[X_1 - X'_1]$ contains a K_{6m-1} . Recall $G[X] = K_{3m-1}$. Then, $G[X \cup X_1]$ contains $2mK_3$ (as $m \geq 2$). Since all vertices of $\otimes_{2 \leq i \leq 3} G^i$ are adjacent to vertices in $X \cup X_1$, G contains a mF^3 , a contradiction. Hence, $X_i - X'_i = \emptyset$ for any $1 \leq i \leq 3$. This means that all the vertices in S_i are isolated in $G[S_i]$ for any

$1 \leq i \leq 3$. Then, G is a spanning subgraph of $K_{3m-1} \otimes T(n - 3m + 1, 3)$. Hence, $G = K_{3m-1} \otimes T(n - 3m + 1, 3)$, as desired. This completes the proof. $\square$

7 Proof of Theorem 1.7

In this section, we give the proof of Theorem 1.7.

Proof of Theorem 1.7. Recall that $\ell \geq 3, m \geq 2\ell$, and $\mathcal{F} = \{F_1, F_2, F_3, F_4\}$, where

$$F_1 = (P_\ell \cup \overline{K_m}) \otimes T(m(r-1), r-1),$$

$$F_2 = mK_2 \otimes mK_2 \otimes T(m(r-2), r-2),$$

$$F_3 = (2K_{\ell-1} \cup \overline{K_m}) \otimes (K_2 \cup \overline{K_m}) \otimes T(m(r-2), r-2),$$

$$F_4 = (K_{\ell-1} \cup \overline{K_m}) \otimes (K_{\ell-1} \cup \overline{K_m}) \otimes T(m(r-2), r-2).$$

First, we can check that for any $1 \leq i \leq r$, F_i contains a P_ℓ after deleting any $(r-1)$ independent sets. Thus, $H \otimes T(n - |H|, r-1)$ is $\mathcal{F}$ -free, if H is P_ℓ -free. For any $n \geq 2mr$, let $\mathcal{G}_n$ be the set of graphs of order n and of form $H \otimes T(n - |H|, r-1)$, where H is a P_ℓ -free graph. Let G_n an extremal graph with maximum edges in $\mathcal{G}_n$. By Lemma 3.4, we can always choose G_n with $H = tK_{\ell-1} \cup K_{\ell_0}$, where $|H| = (\ell-1)t + \ell_0$ with $0 \leq \ell_0 < \ell-1$. Clearly, $0 < |H| - \frac{n}{r} < r\ell$.

Let G' be the graph obtained from G_n by deleting the vertices of $K_{\ell-1}$ in H , and deleting $\ell-1$ vertices in each part of $T(n - |H|, r-1)$ in H . Clearly, $G' \in \mathcal{G}_{n-r(\ell-1)}$ and $e(G') = e(G_n) - (r-1)(\ell-1)(n - (\ell-1)r) - e(T((\ell-1)r, r)) - \frac{\ell-2}{2}(\ell-1)$. Conversely, from $G_{n-r(\ell-1)}$ we can obtain a graph $G'' \in \mathcal{G}_n$ by adding a $K_{\ell-1}$ in its first part, and adding $\ell-1$ independent vertices in each of the remained $r-1$ parts. Clearly, $e(G_{n-r(\ell-1)}) = e(G'') - (r-1)(\ell-1)(n - (\ell-1)r) - e(T((\ell-1)r, r)) - \frac{\ell-2}{2}(\ell-1)$. Since $e(G') \leq e(G_{n-r(\ell-1)})$ and $e(G'') \leq e(G_n)$, we have $e(G') = e(G_{n-r(\ell-1)}) = e(G_n) - (r-1)(\ell-1)(n - (\ell-1)r) - e(T((\ell-1)r, r)) - \frac{\ell-2}{2}(\ell-1)$. Similarly, we can show that for any constant N with $(\ell-1)|N$, $e(G_n) = e(G_{n-Nr}) + e(T(Nr, r)) + (n - rN)(r-1)N + \frac{\ell-2}{2}N$.

For $n \geq 1$, let $\mathfrak{U}_n = \text{EX}(n, \mathcal{F})$. For a graph $G''' \in \text{EX}(n, \mathcal{F})$, we say G''' has property B , if $G''' \in \mathcal{G}_n$. For a graph $G \in \text{EX}(n, \mathcal{F})$, define

$$\phi(G) = e(G) - e(G_n).$$

Clearly, $\phi(G)$ is a non-negative integer-valued function. If G has property B , then $\phi(G) = 0$. By Lemma 3.5, it suffices to show that for sufficiently large n , either G satisfies property B , or there is a $\frac{n}{2} \leq n' < n$ and a $G_0 \in \text{EX}(n', \mathcal{F})$ such that $\phi(G_0) > \phi(G)$.

Since $F_1, F_2 \in \mathcal{F}$, we see that $\mathcal{F}$ satisfies the condition in Lemma 1.2. Since $G \in \text{EX}(n, \mathcal{F})$ with sufficiently large n , there is a constant N (by letting $N_0 = 3m(\ell-1)$ and

$N = O(r, t) \cdot N_0$ in Lemma 1.2), such that G has an induced subgraph $\otimes_{1 \leq i \leq r} G^i$ satisfying:

- (1) $|G^i| = N$ for any $1 \leq i \leq r$;
- (2) for any $1 \leq i \leq r$, $G^i = \cup_{1 \leq j \leq \ell_i} H_j^i$, where $H_1^i, H_2^i, \dots, H_{\ell_i}^i$ are symmetric subgraphs of G and $\ell_i \geq 3m$.

Note that $3m(\ell - 1)|N$. Since G is F_2 -free, there is at most one i , such that $|H_1^i| \geq 2$. Without loss of generality, we can assume that $|H_1^i| = 1$ for $i \geq 2$. That is $G_i = N \cdot K_1$ for $i \geq 2$. Let $V' = V(G) - V(\otimes_{1 \leq i \leq r} G^i)$. By symmetry, we can divide the vertices in V' by $r + 1$ parts as follows. Let X_1 be the set of vertices u in V' , such that u is adjacent to all the vertices of $\otimes_{2 \leq i \leq r} G^i$. For $2 \leq i \leq r$, let X_i be the set of vertices u in V' , such that u is adjacent to all the vertices of $\otimes_{1 \leq j \neq i \leq r} G^j$ but to no vertex of $V(G^i)$. Let $Y = V' - \cup_{1 \leq i \leq r} X_i$, i.e., the set of vertices which have non-neighbors in at least two among G^i with $1 \leq i \leq r$. Note that each vertex in $\cup_{2 \leq i \leq r} X_i$ is adjacent to $(r - 1)N$ vertices of $\otimes_{1 \leq i \leq r} G^i$. Each vertex in Y is adjacent to at most $(r - 1)N - 1$ vertices of $\otimes_{1 \leq i \leq r} G^i$ (using symmetry). The way connecting the vertices in X_1 to $V(G^1)$ is not known now.

Recall that $e(G_n) = e(G_{n-rN}) + e(T(Nr, r)) + (n - rN)(r - 1)N + \frac{\ell - 2}{2}N$. Let $G_0 \in \text{EX}(n - rN, \mathcal{F})$. Then $e(G[V']) \leq e(G_0)$. Note that $\phi(G_0) = e(G_0) - e(G_{n-rN})$. Thus,

$$\begin{aligned} \phi(G) &= e(G) - e(G_n) \\ &= e(G[V']) + e_G(V(\otimes_{1 \leq i \leq r} G^i), V') + e(T(rN, r)) + e(G^1) \\ &\quad - (e(G_{n-rN}) + e(T(Nr, r)) + (n - rN)(r - 1)N + \frac{\ell - 2}{2}N) \\ &\leq \phi(G_0) + e_G(V(\otimes_{1 \leq i \leq r} G^i), V') + e(G^1) - \frac{\ell - 2}{2}N - (n - rN)(r - 1)N. \end{aligned}$$

If $e_G(V(\otimes_{1 \leq i \leq r} G^i), V') + e(G^1) - \frac{\ell - 2}{2}N - (n - rN)(r - 1)N < 0$, then $\phi(G) < \phi(G_0)$, as desired. Thus, we can assume that

$$e_G(V(\otimes_{1 \leq i \leq r} G^i), V') + e(G^1) \geq \frac{\ell - 2}{2}N + (n - rN)(r - 1)N.$$

We will show that G has property B (i.e., $G \in \mathcal{G}_n$).

Similar to Theorem 1.5, we can show that $e_G(V(\otimes_{1 \leq i \leq r} G^i), V') = e_G(V(G^1), X_1) + (r - 1)N \cdot (n - rN) - |Y|$. Recall $e_G(V(\otimes_{1 \leq i \leq r} G^i), V') + e(G^1) \geq \frac{\ell - 2}{2}N + (n - rN)(r - 1)N$. It follows that

$$e(G^1) + e_G(V(G^1), X_1) \geq \frac{\ell - 2}{2}N + |Y| \geq \frac{\ell - 2}{2}N.$$

Since G is F_1 -free, we have $G[X_1 \cup V(G^1)]$ is P_ℓ -free. By Lemma 3.7, we have $e(G^1) + e_G(V(G^1), X_1) = \frac{\ell - 2}{2}N$ (implying $|Y| = 0$). Moreover, either $Q_i^1 = K_{\ell - 1}$ for any $1 \leq i \leq \ell_1$ and $e_G(V(G^1), X_1) = 0$, or (ℓ is even) $Q_i^1 = K_1$ for any $1 \leq i \leq \ell_1$ and there are $\frac{\ell - 2}{2}$ vertices in X_1 which are all adjacent to the (unique) vertex in Q_i^1 for all $1 \leq i \leq \ell_1$. Recall that $Y = \emptyset$. For any $1 \leq i \leq r$, each vertex in $V(Q^i)$ is adjacent to all the vertices in

$\cup_{1 \leq j \neq i \leq r} (X_j \cup V(G^j))$. If $Q_i^1 = K_{\ell-1}$ for any $1 \leq i \leq \ell_1$, then $G[X_i \cup V(G^i)]$ has no edges for any $2 \leq i \leq r$ (otherwise F_3 will be contained in G , a contradiction). Thus, $G \in \mathcal{G}_n$, as desired. It remains to consider the latter case. That is, (ℓ is even) $G^1 = N \cdot K_1$, and there are $\frac{\ell-2}{2}$ vertices in X_1 , say $y_1, y_2, \dots, y_{\frac{\ell-2}{2}}$, such that they are all adjacent to all the vertices of G^1 . Let $X = \{y_1, y_2, \dots, y_{\frac{\ell-2}{2}}\}$. For convenience, we still use X_1 to denote $X_1 - X$. For each $1 \leq i \leq r$, let $S_i = V(G^i) \cup X_i$. Then G has a partition $V(G) = X \cup (\cup_{1 \leq i \leq r} S_i)$ satisfying:

- (3) $X = \{y_1, y_2, \dots, y_{\frac{\ell-2}{2}}\}$ and $S_i = V(G^i) \cup X_i$ for any $1 \leq i \leq r$, where $|V(G^i)| = N \geq 3m$;
- (4) for any $1 \leq i \leq r$, $N_G(v) = V(G) - S_i$ for any $v \in V(G^i)$.

For any $1 \leq i \leq r$, let X'_i be the set of vertices in X_i which have neighbors in X . Similar to Theorem 1.5, we can show that the vertices in X'_i are isolated in $G[S_i]$ for any $1 \leq i \leq r$. Recall that $G[X_i - X'_i]$ is P_ℓ -free for any $1 \leq i \leq r$. By Lemma 3.3, we have $e(G[X_i - X'_i]) \leq \frac{\ell-2}{2}|X_i - X'_i|$ with equality if and only if $G[X_i - X'_i]$ is the disjoint union of several $K_{\ell-1}$. Let $\bar{e}$ be the number of non-edges between two distinct S_i and S_j . Thus,

$$\begin{aligned} e(G) &\leq e(K_{\frac{\ell-2}{2}} \otimes K_{|S_1|, |S_2|, \dots, |S_r|}) + (\sum_{1 \leq i \leq r} e(G[X_i - X'_i])) - (\sum_{1 \leq i \leq r} \frac{\ell-2}{2}|X_i - X'_i|) - \bar{e} \\ &\leq e(K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)) + (\sum_{1 \leq i \leq r} e(G[X_i - X'_i])) - (\sum_{1 \leq i \leq r} \frac{\ell-2}{2}|X_i - X'_i|) - \bar{e} \\ &\leq e(K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r)) - \bar{e}. \end{aligned}$$

However, $e(G) \geq e(K_{\frac{\ell-2}{2}} \otimes T(n - \frac{\ell-2}{2}, r))$ as $G \in \text{EX}(n, \mathcal{F})$. This forces that $\bar{e} = 0$ and $e(G[X_i - X'_i]) = \frac{\ell-2}{2}|X_i - X'_i|$ for any $1 \leq i \leq r$. Then $G[X_i - X'_i]$ is the disjoint union of several $K_{\ell-1}$. If $X_i - X'_i$ is not empty for two i , say $i = 1, 2$ without loss of generality, then both $G[X_1 - X'_1]$ and $G[X_2 - X'_2]$ contains a $K_{\ell-1}$. Since $\bar{e} = 0$, there are no non-edges between these two $K_{\ell-1}$. Thus, F_4 is contained in G , a contradiction. Hence, $X_i - X'_i \neq \emptyset$ for at most one i . Without loss of generality, we can assume that $X_i - X'_i = \emptyset$ for any $2 \leq i \leq r$. This means that all the vertices in S_i are isolated in $G[S_i]$ for any $2 \leq i \leq r$. Note that $G[X \cup S_1]$ is P_ℓ -free, since G is F_1 -free. Then, $G \in \mathcal{G}_n$, as desired. This completes the proof. $\square$

Declaration of competing interest

There is no conflict of interest.

Data availability statement

No data was used for the research described in the article.

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